CoRTeC: A Differentially Private Release for Synthetic Tabular Data, Decodable by Any Generator

Calister Nnona
calisternnona@gmail.com
DOI: 10.5281/zenodo.22945315
Version 1.2.5, 30 October 2026

Abstract. Differentially private synthetic tabular data is produced today by marginal mechanisms, which choose their workload on the private rows and ship a fitted model, or by language models fine-tuned or driven under the budget, inseparable from the mechanism that spends it. We propose CoRTeC, which separates the two. CoRTeC spends the privacy budget once, on a small release of noisy counts built for training a classifier. For each cohort of the data it publishes the cohort's size, its class balance, a histogram of every attribute for each outcome class, and the outcome rate in a set of non-overlapping cells. A reviewer can check every number before the release leaves the institution. Everything after the release is post-processing and costs no further privacy. A maximum-entropy decoder turns the counts into rows, five frozen language models can do the same without seeing a record, and a selection step matches any decoder's rows to the released counts. One release therefore serves any number of draws, by any team and with any model, and the model can be changed without touching the guarantee. At $\varepsilon = 2$, with 300 and 1,000 rows, the auto-configured release decoded without a language model trains classifiers within 0.02 AUC of a real sample of the same size on Adult, over three splits. The gap is larger on NHANES and bank marketing, up to 0.07 AUC on the medical dataset. Its utility reaches that of the reference AIM in five of the six dataset-size settings, while AIM's adaptive workload fits pairs of columns more closely. Four controls show what each part contributes. What the language model adds comes from the column names, and without the release it copies the public file. No private row reaches any decoder, the strongest membership attack stays far below the bound the budget allows, and no synthetic row copies a training row.

1. Introduction

An institution that holds sensitive tabular data, such as a hospital, an insurer or a statistics office, often wants to train models outside its own walls, give realistic samples to partners, and let researchers study a problem without studying the people in the data. Differentially private synthetic data [14, 15] is the accepted way to do all three. Two families of methods are used for this. Marginal methods such as MST [27] and AIM [28] measure low-order marginals under the privacy budget and fit a graphical model to them. A recent benchmark finds that they give the most accurate synthetic data of the methods it tests, on five datasets [10]. Methods built on language models either fine-tune a pretrained model on the private rows under DP-SGD [1, 3, 38], or drive a model through its inference interface and score its candidates privately [24, 43]. Tab-PE keeps that private scoring and replaces the model with variation operators [39]. The benchmark leaves the language-model family out on purpose, because a model pretrained on public text may already know the public benchmarks it is tested on [10]. Swanberg et al. find that rows written by a strong model from the column names alone, used as public data, help no more than uniform data on Adult [37].

Each of these methods spends the budget as it goes. The marginal methods choose their marginals one round at a time on the private rows, and they release a fitted model whose structure reflects those choices, so a new workload means a new run on the private data. The fine-tuning methods put the private rows on the training hardware and spend budget on every gradient step, so a new model means a new training run. The inference-interface methods spend budget on every round of candidates. These designs are built for other goals. None of them is meant to let an institution spend the budget once, on counts chosen in advance, and decode those counts again later, by another team or with another model, at no further cost. Nor can the language model's contribution be measured apart from the private statistics, because the model is part of the mechanism that spends the budget.

In this paper we propose CoRTeC, a differentially private release built for training a downstream classifier, together with two ways of decoding it (Figure 1). Stage A is the only code that reads the private data. It groups the rows into cohorts and, for each cohort, publishes four things with Laplace noise: the cohort's size, its class balance, a histogram of every attribute for each outcome class, and the outcome rate in each of a set of non-overlapping cells. The workload is fixed before the release is computed, either by a public rule or by one selection step whose cost is charged to the budget and whose output is itself released (Section 4.1). The budget is spent there, once. The release leaves the institution as a file of counts that anyone can inspect cell by cell, and everything after it is post-processing.

A maximum-entropy decoder, the same estimator AIM uses, fits the released counts and samples rows from them. A frozen language model can decode the same release instead. It reads the release and nothing else, and writes rows that carry exactly the counts it is asked for. A selection step then takes any decoder's rows, matches them to the released counts, and redraws values inside their released bins. Because every reader is post-processing, generation can be repeated without limit and on any model, including one behind an interface the institution does not control. The model can then be swapped for another at no privacy cost.

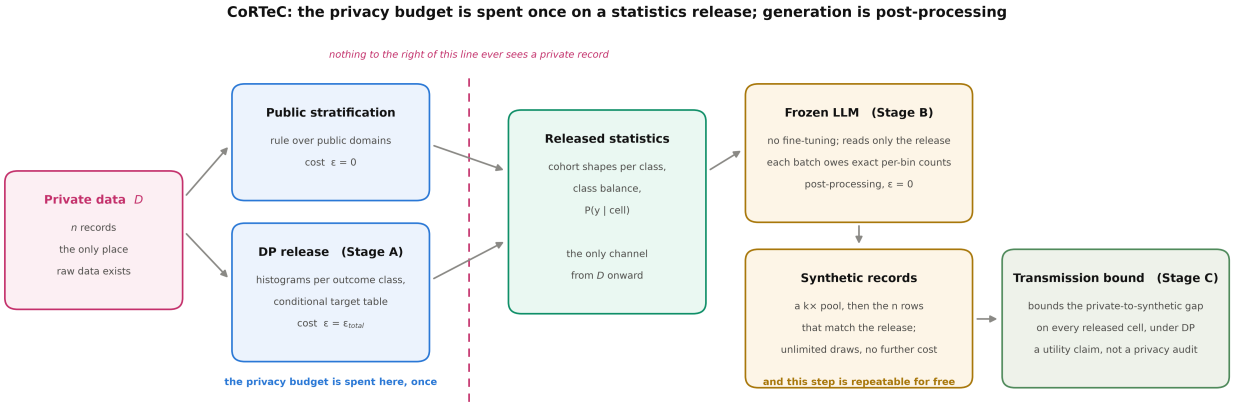


Figure 1: The privacy budget is spent once, in Stage A. Everything to the right of the dashed line reads only released statistics, so it can be repeated without limit.

What an institution gains. The design gives an institution five things. First, the whole privacy loss is incurred by one file, in which every number records its query, its budget and how it composes. The review before a release is therefore a review of counts, not of model weights or a training run. Second, the workload is set before the release is computed, so the conditional rates a deployment needs are declared up front rather than discovered round by round. Third, only the release crosses the institution's boundary. The rows never reach a model, a training cluster or a vendor, and whoever runs the decoder, even a hostile endpoint, learns nothing the release does not say. Fourth, decoding is free and repeatable. Another team can decode the same release at another size, with a later model or with no model at all, and a decoder that disappoints can be replaced without touching the guarantee. Fifth, because the model is separate from the mechanism, its contribution can be measured, and we use that to test our own claims.

We evaluate CoRTeC on three datasets: Adult, NHANES and bank marketing. As reference points we run the reference implementations of AIM and MST on the same split and public bins, and DP-2Stage [3], DP-LLMTGen [38] and Tab-PE [39] under each paper's own protocol. Four controls separate the parts of the design: the release decoded without a language model, alone and with the selection step; the schema renamed to neutral tokens; and rows written from the column names with no release. Every table reports fidelity and utility side by side, and the central results are repeated on three splits.

The release on its own is enough to train useful models. Auto-configured and decoded without a language model, it trains every student classifier to within 0.02 AUC of a real sample of the same size on Adult, over three splits. Across the three datasets its utility reaches that of the reference AIM in five of the six dataset-size settings. With the fixed cohort rule, it carries the conditional rates it names on Adult more closely than AIM's draws do, and AIM's adaptive workload fits pairs of columns more closely at 1,000 rows on Adult and at both sizes on NHANES. The gap to a real sample is largest on the smallest dataset, the medical survey, where it reaches 0.07 AUC at 300 rows. What the language model adds comes from the column names. It repairs the pairs of columns it recognises, invents links between columns that are nearly independent, and loses its advantage when the names are replaced by neutral tokens. Without the release it copies the public file into most of its output. In every arm no private row reaches the model and no synthetic row copies a training row.

Our contributions are as follows.

- **The release** (Section 4.1). One file of noisy counts carries the whole privacy loss. It holds class-conditional histograms and a conditional outcome table inside cohorts, with the workload fixed before the release is computed. It is released under pure ε, with a five-step privacy proof, and can be inspected cell by cell. An auto-configurator chooses the cohorts and the conditional levels within the same budget.

- **Two decoders and a selection step** (Section 4.2). A maximum-entropy decoder that rebuilds rows at no cost, a frozen language model guided by exact per-bin counts, and Algorithm 2, which matches any decoder's output to the released counts. The selection step accounts for most of the gain over the plain decoder.
- **A reference implementation** that ships both decoders (`cortec run`, `cortec decode`) and the checks a deployment needs: membership inference tested on a positive control, distance to the closest record, a memorisation test of the model, and the cost per thousand rows.
- **An evaluation** (Section 5) that reports fidelity and utility together, with floors and a ceiling in every table and replication over three splits. It covers the reference AIM and MST, three recent methods under their own protocols, and four controls that assign each gain to the release, the selection step or the model.

A companion deployment guide covers the practical side: the reference architecture, the guardrails, and the failure modes we met while building the method, each with the test that catches it.

2. Related Work

Three families of methods are in wide use. Some measure marginals under the budget, some train a generator under the budget, and some keep a foundation model frozen. Section 5 runs the members of each family that an institution would consider: the reference AIM and MST, DP-2Stage, the DP-LLMTGen procedure on three backbones, and Tab-PE.

Marginal methods. PrivBayes [46], MST [27], PrivMRF [7] and AIM [28] select low-order marginals under the budget and fit a graphical model with Private-PGM [26]. MST took first place in the 2018 NIST challenge, whose entries Bowen and Snoke compare on marginal, joint and correlation measures [6]. AIM chooses its marginals adaptively, and it is our main reference from this family. The benchmark of Chen et al. [10] finds AIM and PrivMRF the most accurate over five datasets and three budgets, measured by machine-learning efficacy, 3-way query error and 2-way total variation. The benchmark of Tao et al. [36] reached a similar conclusion for the generative methods of its time. Our reference AIM and MST rows use the authors' private-pgm implementation on our public bins.

Generators trained under differential privacy. DP-GAN [44] and DP-CTGAN [32] train a generative adversarial network under DP-SGD [1], the second one based on CTGAN [42]. PATE-GAN [22] trains one under PATE [30, 31]. DP-TBART [8] trains an autoregressive transformer under DP-SGD and matches the marginal methods on most of its datasets, and DP-TLDM [47] trains a latent diffusion model in the same way. Those papers and the benchmark evaluate these generators alongside the marginal methods [3, 10, 38]. We cite them here, and run the methods whose design is closest to CoRTeC's.

Language models fine-tuned on the private rows. GReaT [5] writes each row as "column is value" text and fine-tunes GPT-2, with no privacy guarantee. DP-LLMTGen [38] fine-tunes Llama-2 7B in two stages. It first learns the table format on random tables, then trains on the private rows under DP-SGD with a loss that weights the value tokens. Its authors find that meaningful feature names help the fine-tuned model. DP-2Stage [3] fine-tunes GPT-2 in the same way, with a pseudo-data stage first, and includes AIM among its reference points on Adult. In all of these methods the private rows reach the training hardware, and the budget is spent on per-example gradients in every epoch.

Methods that keep the model frozen. Private Evolution [24] and its text version [43] generate candidates from a foundation model, let the private records vote for their nearest candidates under a Gaussian mechanism, and repeat. Tab-PE [39] replaces the model with simple variation operators, and shows that marginal methods miss high-order correlations that the standard benchmarks do not test. Swanberg et al. [37] adapt Private Evolution to tabular data. They also try rows generated in one shot from the column names, used as public data inside a marginal mechanism. On Adult they find that neither approach improves on the marginal baselines, and that the one-shot rows reproduce single columns but not pairs [37]. We run Tab-PE [39] and the one-shot arm as well. CoRTeC differs from both in what the model is given: a differentially private release of conditional statistics, which the selection step then enforces. Table 1 summarises the differences.

Table 1: What each method needs. The budget column says where the privacy loss is incurred.

method	model weights	private rows seen by	public or side data	budget spent on	adaptive rounds	backbone swap
CoRTeC	no; any inference interface	Stage A only; the model never sees a row	column names and value lists	one release of counts, composed once	none	at no privacy cost
AIM, MST	none	the measured marginals	public domains and bins	measured marginals; AIM selects adaptively	AIM yes	not applicable
DP-2Stage	GPT-2, full fine-tune	the DP-SGD trainer	pseudo data (uniform over private ranges, or a public table)	per-sample gradients, every step	yes	retrain
DP-LLMTGen	Llama-2 7B, LoRA	the DP-SGD trainer	random tables from names, categories and ranges	per-sample gradients, every step	yes	retrain
Tab-PE	none	the nearest-neighbour histogram, every iteration	numerical bounds; class counts	Gaussian noise on the histogram, 15 to 30 rounds	yes	not applicable
one-shot records [37]	no	none by the model; the downstream mechanism	column names and types	inside the downstream mechanism	depends	at no privacy cost

3. Preliminaries and Privacy Model

A dataset D is a multiset of n records over attributes $A = A_{num} \cup A_{cat}$ and a binary target y. Each attribute has a public domain: an interval for a numerical attribute and a finite set of values for a categorical one. Public bin edges divide each numerical domain. $M_a(D)$ is the histogram of attribute a over its bins. For a cell c of the record space, $k_c(D)$ is the number of records in c with $y = 1$. Two datasets are neighbours if one can be obtained from the other by adding or removing one record.

Definition 1 (Differential privacy [14]). *A randomised mechanism M is ε-differentially private if for all neighbours D, D' and all sets S of outputs, $Pr[M(D) \in S] \le e^{\varepsilon} Pr[M(D') \in S]$.*

The Laplace mechanism adds $\mathrm{Lap}(\Delta/\varepsilon)$ noise to a function of L1 sensitivity Δ and is ε-differentially private [14]. A histogram over any number of bins has $\Delta = 1$, and so does a counting query. Mechanisms run on the same data compose sequentially, so their budgets add [15]. Mechanisms run on disjoint parts of the data, defined by a rule that does not depend on the data, compose in parallel and cost only the largest of their budgets [29]. Any computation on the output of a private mechanism that does not read D again is post-processing and costs nothing [15]. Definition 1 also limits any membership attacker. No test of whether a record is in D can have a true-positive rate that exceeds its false-positive rate by more than $(e^{\varepsilon} - 1)/(e^{\varepsilon} + 1)$, which is 0.762 at $\varepsilon = 2$.

Threat model. The adversary has unlimited computing power and sees everything that crosses the trust boundary: the release R, every synthetic dataset drawn from it, and the model's inputs and outputs. The institution's data extract, the schema and the release engine run inside its perimeter, and they are the only code that reads D. The generator, the selection step and every later use read only R and their own output. Whoever operates the model, including a hostile endpoint, therefore learns nothing that R does not reveal, and R is ε-differentially private.

Privacy model. The guarantee is pure ε ($\delta = 0$), central, under add/remove adjacency, and it protects one row. On data with one row per person it protects a person. On data with several rows per person, such as hospital encounters, the per-person guarantee weakens by the largest number of rows any one person contributes, and the reference implementation requires that number to be declared. As in every marginal method, two things are treated as public: the list of attributes with their domains and bins, and the rule that forms the cohorts. Cohort sizes, class balances and cell sizes are private counts, so they are released with noise rather than assumed. The model's pretraining corpus is outside the guarantee. If a private record was in it, that record was exposed before CoRTeC ran. Like prior work, we treat a public pretrained model as public [37, 40], and we measure the consequence that matters: whether anything the model memorised reaches a draw (Section 5.5).

4. CoRTeC: the Release and its Decoders

CoRTeC has three stages (Figure 1). Stage A is the only code that reads D, and it produces the release R. Stage B decodes R, either with a maximum-entropy model or with a frozen language model, and passes the decoder's output through a selection step. It reads only R and its own output. Stage C is optional and has its own budget. It bounds how

far the synthetic conditional rates are from the private ones. This section describes Stage A and its design choices, walks through one cohort of a stored release, describes the decoders and the selection step, and states the privacy guarantee.

4.1 Stage A: the release

Algorithm 1: CoRTeC-Release

Input: private D; public schema (domains, bin edges, stratification rule Π, conditional levels $L_0 \subset \ldots \subset L_k$); budget ε_{total}; conditional share α; count share γ; suppression floor n_{min}

Output: release R

(1) $\varepsilon_{count} \leftarrow \gamma\, \varepsilon_{total}$; $\varepsilon_{marg} \leftarrow (1 - \alpha)(\varepsilon_{total} - \varepsilon_{count})$; $\varepsilon_{cond} \leftarrow \alpha\, (\varepsilon_{total} - \varepsilon_{count})$

(2) $q \leftarrow |A| + 1$ queries per cohort; $\varepsilon_q \leftarrow \varepsilon_{marg} / q$; $\varepsilon_L \leftarrow \varepsilon_{cond} / |L|$

(3) For each cohort P the rule Π can form, holding records or not:

  (4) $\tilde{n_P} \leftarrow |P| + \text{Lap}(1/\varepsilon_{count})$; if $\tilde{n_P} < n_{min}$ continue; release $\tilde{n_P}$

  (5) release the two-bin histogram of y on P, each count + $\text{Lap}(1/\varepsilon_q)$

  (6) if both noised class counts are at least n_{min}: for $c \in \{0, 1\}$ and $a \in A$ release $M_a(P \cap \{y = c\}) + \text{Lap}(1/\varepsilon_q)$

  (7) else: for $a \in A$ release $M_a(P) + \text{Lap}(1/\varepsilon_q)$ (pooled block)

(8) For each level $\ell \in L$ and each cell c of its public domain, holding records or not:

  (9) $\tilde{k_c} \leftarrow k_c(D) + \text{Lap}(2/\varepsilon_L)$; $\tilde{n_c} \leftarrow |c| + \text{Lap}(2/\varepsilon_L)$; if $\tilde{n_c} < n_{min}$ continue

  (10) release $\tilde{\rho_c} \leftarrow \text{clip}(\tilde{k_c} / \max(\tilde{n_c}, n_{min}), 0, 1)$

(11) Return R with every released quantity, its ε, sensitivity and composition rule

Four design choices keep the release cheap. First, the cohorts come from a rule over public domains, so forming them costs nothing. The cohorts do not overlap, so their queries compose in parallel. Second, the release uses histograms instead of means and standard deviations. A histogram has sensitivity 1 however many bins it has, so the whole shape of a distribution costs the same as a single count. Third, a separate histogram block for each outcome class costs no more than one pooled block, because the two class blocks do not overlap. The pooled histogram is their mixture under the released class balance, which is post-processing. A cohort whose smaller class falls below n_{min} keeps the pooled block instead. Fourth, the conditional table is released over the cells of a partition, so each level receives its full budget ε_L however many cells it holds. Each cell's rate is the ratio of two counts whose noise does not depend on anything private, which keeps the mechanism pure ε. Noising the rate directly would need noise scaled to the private cell size, and that would not be pure ε. Every decision about n_{min} compares a noised, charged count with the public floor, so which cohorts and cells appear in R depends only on released quantities. An auto-configurator chooses Π and L from the schema, the budget and the requested size. Its one step that reads D, a noised contingency table for each candidate column, is charged to the budget (Appendix D).

Example 1 (one cohort of a release). Take the seed-42 Adult release at $\varepsilon = 2$. Every fixed-rule Adult arm of Section 5 decodes this release, the anonymised arms under neutral names. It forms twelve cohorts from four education bands and three bands of weekly hours. Consider the cohort of graduates who work forty hours a week. With $\gamma = 0.02$ the cohort-size query has $\varepsilon_{count} = 0.04$, so Stage A releases the true size plus Laplace noise at scale 25. The released size is 762. The class-balance query and the fourteen histograms share $\varepsilon_{marg} = 1.568$, as fifteen queries at $\varepsilon_q = 0.105$ each, with noise at scale 9.6 on every count. The two noised class counts give a balance of 0.551 above 50K and 0.449 at or below. Both counts are above $n_{min} = 150$, so the cohort receives one histogram block per class. In the block above 50K, age puts 0.216 of its mass in the 40 to 45 bin and none below 25, and relationship is 0.744 husband and 0.179 not in a family. In the block at or below 50K, age puts 0.140 in the 25 to 30 bin and 0.049 below 25, and relationship is 0.282 husband and 0.419 not in a family. A category whose noised count fell below the floor is listed as suppressed and gets weight zero in the selection step. Here that is other relative, own child and unmarried above 50K, and other relative at or below. The two levels of the conditional table share $\varepsilon_{cond} = 0.392$. Level 0 is a single cell holding the overall rate, 0.241. Level 3 crosses education, hours and marital status, and gives this cohort two cells: 0.739 for married people and 0.210 for the rest. Each rate is the ratio of two counts noised at scale 10.2, taken over the public floor. That is everything about this cohort that leaves the institution: a size, two class counts, twenty-eight histograms, a list of suppressed categories and two rates, each with its ε recorded beside it. Stage B at $n = 300$ with a pool factor of 3 gives this cohort 26 pool rows, in proportion to its released size. The released balance assigns 14 of them to the class above 50K and 12 to the class below. Each class block then sets how many rows each bin must hold, by the largest-remainder rule. Above 50K that means three rows aged 40 to 45 and ten husbands. Below 50K it means two

rows aged 25 to 30 and five people not in a family. The prompt states these counts, and the language model writes rows that carry them. The maximum-entropy decoder has no prompt and samples its fitted distribution at the same pool size. The selection step then matches the pool of every cohort to the released counts, keeps about nine rows of this cohort in the final 300, and refines them by swapping rows. Finally it redraws each numerical value inside its released bin, so the rows that leave carry the release's structure down to bin level and nothing of the model's below it. Figure 2 shows the two class blocks of this cohort as released.

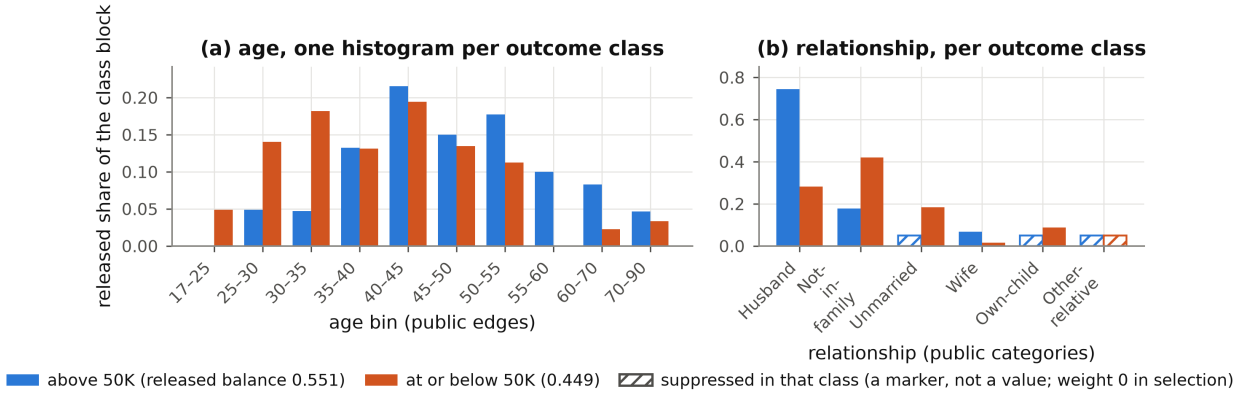


Figure 2: The cohort of Example 1 as it leaves the institution. (a) The age histogram released for each outcome class, over the public bin edges. (b) The relationship shares for each class. Hatched markers show categories whose noised count fell below the floor, which get weight zero in the selection step. Every bar is a released number. The decoders read these numbers, the class balance, the cohort size and the two conditional cells, and nothing else.

4.2 Stage B: exact counts, selection and the sub-bin rule

Stage B divides the requested rows among the cohorts in proportion to their released sizes. From the released histograms it works out exact counts for every bin, category and class, and asks the frozen model for batches of 25 rows that carry the counts the cohort still owes. The prompt (Appendix D) contains the release and the counts and never a record. It states the counts separately for each outcome class, and it tells the model to follow the released statistics even where they contradict its expectations. Rows that parse, lie inside their domains and use declared categories enter a pool k times the requested size, with $k = 3$ on Adult and bank and 2 on NHANES. Generation runs per cohort, not per released cell. A cell-by-cell variant produced nothing for the people the released cells did not name, and dropped whole subgroups while every aggregate check passed. This is the kind of disparate impact Ganev et al. document for differentially private synthesis [17], and the reference implementation refuses that path below a coverage floor (deployment guide).

Algorithm 2: CoRTeC-Select

Input: release R; pool $D\hat{}$; requested size m

Output: synthetic dataset D^* of m rows

(1) Require that every cohort P holds at least $m\, n\tilde{}_P / \Sigma\, n\tilde{}$ pool rows and a tenth more (coverage guard)

(2) $w \leftarrow$ inclusion weights by iterative proportional fitting [12] of $D\hat{}$ to every released cell: cohort sizes, class balances and every histogram; a suppressed category receives weight 0

(3) $D^* \leftarrow$ a systematic sample of m rows under w (rake)

(4) Swap a row of D^* for a pool row whenever the swap lowers the weighted L1 distance between the cell counts of D^* and R, until no swap improves it (refine)

(5) In every cohort with class blocks, redraw each numerical value uniformly inside its released bin intersected with its stratification band; integer columns draw integers (sub-bin rule)

(6) Return D^*

Exact counts make the pool match the release closely. The selection step then removes the sampling error that any m rows drawn from a faithful pool still carry. The sub-bin rule deals with the structure inside each bin. A released histogram fixes the count in each bin and nothing finer, so where a value falls inside its bin should come from the release, not from the model. Each step reads only R, the generated rows and fresh randomness. Whether a numerical

column holds integers is decided from the public bin edges and bounds, together with the majority of the pool's values. A rule that looked at every value instead let one model-written "45.0" turn a whole column into real-valued redraws. Every draw in this paper is scored with the majority rule in place (Appendix D).

The maximum-entropy decoder. To measure what the language model contributes, we also decode the same release with no model. The decoder fits the maximum-entropy distribution that agrees with every released count, using Private-PGM [26], and samples from it at the same n. It reads only R, so it spends no budget, and every language-model row in Section 5 is read against it. Seen from the side of the marginal methods, the release is a fixed workload: class-conditional single-column marginals inside cohorts, plus the conditional cells, chosen before the release is computed. The decoder fits that workload with the same estimator AIM uses. What separates CoRTeC's release from AIM is therefore the workload, not the inference engine. AIM's adaptive selection gives it better pairs of columns (Section 5). The fixed workload gives the conditional rates a deployment names, a file a reader can inspect cell by cell, and a release computed in seconds.

Stage C. For each released cell, the private data has a positive rate p_c and the synthetic data a public rate q_c. Stage C releases a Laplace estimate of p_c, taken over the public floor, at a declared budget $\varepsilon_{\text{cert}}$. It turns that estimate into a bound on $|p_c - q_c|$ that holds for all cells at once. It gives no verdict unless a real hold-out sample passes the tolerance and the same sample with its target shuffled fails it, and unless every cell holds at least 20 synthetic rows. The bound concerns utility only (Appendix E.8).

4.3 Privacy analysis

Theorem 1. *Algorithms 1 and 2 together with the generation step are ε_{total}-differentially private; with auto-configuration the total is $\varepsilon_{sel} + \varepsilon_{release} = \varepsilon_{total}$; Stage C adds ε_{cert}.*

The proof (Appendix A) has five composition steps. (i) Forming the cohorts reads public domains only and is free. (ii) Within a cohort, the class-balance query and the $|A|$ histograms of a record's own class block form a chain of q Laplace queries of sensitivity 1, which compose sequentially to $\varepsilon_{\text{marg}}$. The two class blocks do not overlap and compose in parallel. (iii) Across cohorts the queries compose in parallel, because each record lies in exactly one cohort. (iv) Within a conditional level the cells partition D and compose in parallel. A cell's two counts compose sequentially to ε_L, and the published rate is post-processing of the two counts and the public floor. (v) The levels describe the same people, so their budgets add up to $\varepsilon_{\text{cond}}$. The cohort sizes cost $\varepsilon_{\text{count}}$, in parallel across cohorts. Generation, selection and the sub-bin rule read only R, the generated rows and randomness, so by post-processing they add nothing, at any pool factor and for any number of draws. Summing gives $\varepsilon_{\text{count}} + \varepsilon_{\text{marg}} + \varepsilon_{\text{cond}} = \varepsilon_{\text{total}}$.

5. Experiments

We evaluate CoRTeC on three public datasets. As reference points we run the reference implementations of AIM and MST, and DP-2Stage, the DP-LLMTGen procedure and Tab-PE, both on our split and under each paper's own protocol. We also run the four controls of Section 1. Utility is the measure a deployment is judged on. We report fidelity beside it in every table, because the rows themselves are also handed out.

5.1 Setup

Table 2: Datasets. Every split uses seed 42. The replication seeds 43 and 44 each draw a fresh split and a fresh release.

dataset	records	train / test	numerical / categorical	target, positive rate	role
UCI Adult [13, 23]	32,561	26,048 / 6,513	6 / 8	income over 50K, 24.1 %	every comparison, the controls, the ladder, the replication
NHANES 2017-2018 [9]	3,749 adults	2,999 / 750	5 / 4	HbA1c at or above 6.5 %, 14.2 %	auto-configured clinical dataset, the replication
bank marketing [25]	45,211	36,168 / 9,043	5 / 8	term deposit, 11.7 %	third dataset, one split

Reference methods. The reference AIM and MST are the authors' private-pgm implementation (mbi 1.1), whose empirical privacy Ganev et al. audit as close to the stated guarantee [16]. We fit both on the training split, with the same public bins CoRTeC declares, at the same ε, and with a corrected sampler. AIM runs under the stable message-passing oracle, with a check that the fitted marginals agree with the noisy measurements. MST, and the AIM and

MST rows at $\varepsilon = 0.5$ and 1 in Appendix E.7, run under the library's default oracle. Appendix D gives the settings, the reason for the oracle and how closely the two oracles agree on the seed-42 split. DP-2Stage runs the authors' code (GPT-2, DP-SGD, $\delta = 10^{-5}$) on our split. The DP fine-tuning rows apply DP-LLMTGen's two-stage LoRA [20] procedure and loss to Qwen3-14B, Qwen2.5-7B and Llama-2 7B on our split ($\delta = 10^{-5}$), so that the same backbone appears under two privacy designs. Tab-PE runs the authors' code with a population of 1,000 rows (600 on NHANES) and $\delta = 1/(n \ln n)$, and its own evaluator, TabICL [19], scores the Tab-PE protocol rows. Two controls complete a two-by-two design over the release and the model. The release without a model is the maximum-entropy decoder of Section 4.2, alone and with the selection step. The release with a model is CoRTeC. The model without a release writes rows from the column names alone, the baseline Swanberg et al. use [37]. The permuted-target floor has neither.

CoRTeC configuration. One set of settings was fixed before the experiments: $\varepsilon = 2$, $\alpha = 0.2$, $\gamma = 0.02$ and $n_{\min} = 150$; batches of 25 rows; counts stated per outcome class; a pool factor of 3 (2 on NHANES); and the selection step with the sub-bin rule. NHANES and bank marketing use the auto-configurator to choose their cohorts and conditional levels, and Adult uses the hand-written rule in its schema. The auto-configurator is the shipped default, so after the main experiments we ran it on Adult as well. We did this for consistency, not as a search over configurations. On each split the released Stage A chooses the cohorts and the hierarchy, charges that choice to the same budget, and spends a count share of 0.05 plus the gate charge instead of 0.02 (Appendix D). This release is decoded with no model and the selection step only, and it is labelled "auto-configured, no model, with selection" wherever it appears. The language-model backbones are Qwen3-8B, Qwen3-14B and Qwen3-32B (4-bit AWQ), served locally through vLLM, and Gemini 3.6 Flash through Google's interface, which has served the Gemini 3.5 Flash identifier since 8 October 2026. Qwen3-14B is the primary arm. A Qwen2.5-7B arm appears only where a protocol was run on it (Appendix E.6).

The privacy budget. Recent papers report $\varepsilon = 1$ as their headline, the benchmark of Chen et al. uses 0.2, 1 and 5, and the adversarial generators are also reported at $\varepsilon = 1$ [3, 38]. Our headline of $\varepsilon = 2$ comes from the release itself. At the shipped split each count of a conditional cell is noised at scale 10.2, so a cell at the floor of 150 records carries its rate to within about 0.1. At $\varepsilon = 1$ and 0.5 that error grows to about 0.2 and 0.4. The conditional table then stops being informative near the floor, and a comparison would measure the noise rather than the methods. We chose $\varepsilon = 2$ as the budget at which a cell of floor size still carries its rate to within a tenth of the range. It is not a measured threshold, and we do not claim that a trade-off at $\varepsilon = 2$ disappears below it. Every method also appears at a budget its own paper uses. The reference AIM, MST and the release run at $\varepsilon = 0.5$ and 1 under the DP-LLMTGen protocol and at $\varepsilon = 1$ under the DP-2Stage and Tab-PE protocols, and the release's own decoder runs at $\varepsilon = 0.5$ and 1 under our protocol (Appendix E.7). The picture is the same at every budget. Under each paper's own protocol on Adult, the release and its decoder reach the utility that paper reports, and the reference AIM fits the private table more closely (Appendix E.6).

Metrics. Fidelity is measured four ways against the full private training table, over the public bins. 1-way and 2-way total variation compare single columns and pairs of columns. Conditional total variation compares $P(y \mid \text{group})$ over two families of groups: the cohort bands the release describes ("seen") and groups built from columns that share nothing with the conditioning ("held-out"). Utility is train-on-synthetic, test-on-real AUC [32]. Logistic regression, a random forest and gradient boosting are trained on the synthetic table and tested on the real test split. Appendix B gives the definitions. Each paper protocol in Appendix E.6 uses that paper's own metrics and split.

Floors, ceiling and repetition. Every table carries three references at the same n. The first is a real sample of n training rows, which a synthetic table of the same size is not expected to exceed. The second is the same rows with the target shuffled, a floor that carries no information. The third is the students trained on a real sample of 4,000 training rows (the whole split on NHANES, which has 2,999), the ceiling. A metric that cannot separate the first two references has no power, and we checked this before reading any table. Measuring against explicit reference points rather than a bare threshold is the practice DPBench recommends [18]. Every cell is the mean over three draws, with the sample standard deviation. The central cells are repeated on seeds 43 and 44 with fresh splits and releases, and the paired differences across seeds are reported with their sign count and a paired t-test. Results from a single split or draw are labelled as such and read as signals only. With three seeds a paired test has little power, so we treat a difference as replicated when its sign is the same at every seed.

5.2 Without a language model, the release trains models close to a real sample

Table 3: Adult at ε = 2 on the seed-42 split. Fidelity is total variation (lower is better) and utility is train-on-synthetic, test-on-real AUC (higher is better). Each cell is the mean ± standard deviation over three draws (five for the reference MST; the ceiling is one fit). The reference rows are in italics. The Gemini arm was run at $n = 300$ only. The decoder-with-selection row is the release decoded with no model and passed through Algorithm 2 on a pool of three times n. The one-shot row is the Gemini backbone writing rows from the column names alone, with no release (Appendix D). The auto-configured row is the release the auto-configurator derives on Adult (Section 5.1), decoded with no model and the selection step; its conditional cells are not comparable (n/a; Appendix E.2). Appendix F gives the membership advantage of every arm.

$n = 300$

condition	1-way TV ↓	2-way TV ↓	cond. seen ↓	cond. held-out ↓	TSTR-LR ↑	TSTR-RF ↑	TSTR-GBM ↑
real sample of 4,000 rows (utility ceiling)	*0.011*	*0.028*	*0.020*	*0.008*	*0.860*	*0.907*	*0.914*
real sample of n rows (floor)	*0.043 ± 0.005*	*0.107 ± 0.004*	*0.062 ± 0.031*	*0.036 ± 0.001*	*0.830 ± 0.013*	*0.870 ± 0.016*	*0.840 ± 0.022*
permuted target (no information)	*0.041 ± 0.002*	*0.110 ± 0.000*	*0.152 ± 0.045*	*0.102 ± 0.013*	*0.423 ± 0.200*	*0.485 ± 0.127*	*0.482 ± 0.107*
CoRTeC, Qwen3-14B	0.046 ± 0.011	0.139 ± 0.010	0.027 ± 0.011	0.048 ± 0.016	0.807 ± 0.026	0.862 ± 0.015	0.833 ± 0.019
CoRTeC, Gemini 3.6 Flash	0.027 ± 0.001	0.110 ± 0.001	0.013 ± 0.008	0.042 ± 0.010	0.805 ± 0.013	0.863 ± 0.008	0.838 ± 0.005
maximum-entropy decoder of the release	0.058 ± 0.001	0.146 ± 0.002	0.082 ± 0.018	0.049 ± 0.011	0.818 ± 0.020	0.856 ± 0.017	0.847 ± 0.009
maximum-entropy decoder with the selection step	0.033 ± 0.011	0.118 ± 0.006	0.014 ± 0.006	0.048 ± 0.011	0.824 ± 0.009	0.866 ± 0.004	0.862 ± 0.006
auto-configured release, no model, with selection	0.019 ± 0.002	0.108 ± 0.001	n/a	n/a	0.826 ± 0.016	0.873 ± 0.009	0.860 ± 0.006
reference AIM	0.056 ± 0.005	0.122 ± 0.001	0.055 ± 0.007	0.047 ± 0.015	0.845 ± 0.004	0.869 ± 0.012	0.845 ± 0.012
reference MST	0.048 ± 0.005	0.121 ± 0.004	0.140 ± 0.021	0.043 ± 0.006	0.733 ± 0.051	0.802 ± 0.021	0.748 ± 0.035
DP-2Stage, GPT-2	0.280 ± 0.006	0.465 ± 0.007	0.147 ± 0.005	0.141 ± 0.020	0.598 ± 0.090	0.607 ± 0.155	0.601 ± 0.205
DP fine-tuned Qwen3-14B	0.106 ± 0.004	0.190 ± 0.002	0.087 ± 0.020	0.071 ± 0.008	0.809 ± 0.025	0.832 ± 0.010	0.815 ± 0.007
Tab-PE	0.203 ± 0.002	0.366 ± 0.004	0.134 ± 0.002	0.049 ± 0.005	0.812 ± 0.026	0.856 ± 0.013	0.839 ± 0.013
one-shot from the column names, Gemini 3.6 Flash (no release)	0.148 ± 0.002	0.307 ± 0.002	0.214 ± 0.014	0.136 ± 0.007	0.795 ± 0.009	0.850 ± 0.001	0.831 ± 0.012

$n = 1{,}000$

condition	1-way TV ↓	2-way TV ↓	cond. seen ↓	cond. held-out ↓	TSTR-LR ↑	TSTR-RF ↑	TSTR-GBM ↑
real sample of n rows (floor)	*0.026 ± 0.004*	*0.061 ± 0.003*	*0.031 ± 0.012*	*0.019 ± 0.002*	*0.855 ± 0.001*	*0.892 ± 0.002*	*0.883 ± 0.004*
permuted target (no information)	*0.025 ± 0.002*	*0.069 ± 0.003*	*0.146 ± 0.015*	*0.095 ± 0.003*	*0.511 ± 0.119*	*0.508 ± 0.085*	*0.513 ± 0.065*
CoRTeC, Qwen3-14B	0.036 ± 0.002	0.108 ± 0.004	0.022 ± 0.002	0.031 ± 0.005	0.827 ± 0.016	0.860 ± 0.007	0.859 ± 0.007
maximum-entropy decoder of the release	0.041 ± 0.003	0.106 ± 0.005	0.034 ± 0.014	0.045 ± 0.009	0.826 ± 0.008	0.870 ± 0.004	0.863 ± 0.011
maximum-entropy decoder with the selection step	0.027 ± 0.004	0.089 ± 0.003	0.008 ± 0.002	0.037 ± 0.003	0.836 ± 0.006	0.873 ± 0.004	0.874 ± 0.007
auto-configured release, no model, with selection	0.016 ± 0.000	0.081 ± 0.001	n/a	n/a	0.841 ± 0.003	0.884 ± 0.002	0.884 ± 0.003
reference AIM	0.029 ± 0.003	0.072 ± 0.003	0.028 ± 0.007	0.030 ± 0.005	0.854 ± 0.002	0.888 ± 0.006	0.883 ± 0.010
reference MST	0.031 ± 0.003	0.083 ± 0.004	0.132 ± 0.017	0.044 ± 0.013	0.788 ± 0.029	0.818 ± 0.017	0.804 ± 0.016
DP-2Stage, GPT-2	0.277 ± 0.005	0.454 ± 0.006	0.138 ± 0.020	0.133 ± 0.007	0.527 ± 0.113	0.485 ± 0.038	0.484 ± 0.033
DP fine-tuned Qwen3-14B	0.100 ± 0.003	0.172 ± 0.003	0.050 ± 0.010	0.054 ± 0.008	0.809 ± 0.033	0.845 ± 0.007	0.836 ± 0.012
Tab-PE	0.196 ± 0.001	0.350 ± 0.002	0.101 ± 0.017	0.031 ± 0.003	0.819 ± 0.011	0.854 ± 0.011	0.843 ± 0.007
one-shot from the column names, Gemini 3.6 Flash (no release)	0.146 ± 0.002	0.305 ± 0.003	0.191 ± 0.009	0.137 ± 0.005	0.807 ± 0.004	0.860 ± 0.007	0.844 ± 0.012

The release on its own trains models close to a real sample. On Adult, the auto-configured release decoded with no model and the selection step trains every student to within 0.02 AUC of a real sample of the same size, over three splits. On NHANES and bank marketing we decoded it with no model and no selection step. There the gap to a real sample is about the size of the reference AIM's: larger than AIM's on NHANES at $n = 300$, within 0.003 of it in two bank cells, and smaller elsewhere. Table 3 gives the seed-42 split and Figure 3 the means over three seeds. Every paired difference quoted below is in Appendix E.2.

On Adult at $n = 1{,}000$, the maximum-entropy decoder and CoRTeC differ by at most 0.005 in either direction on single columns, pairs, the seen conditionals and every student, over the three seeds. None of these differences reaches $p < 0.1$. On the held-out conditionals CoRTeC is better by 0.009 ($p = 0.07$). At $n = 300$ the decoder is worse on the seen conditionals (0 of 3 seeds in its favour, $p = 0.04$) and better on the gradient-boosting student (3 of 3, $p = 0.01$). On NHANES the decoder fits pairs more closely than CoRTeC at every seed, and scores higher on every student at $n = 1{,}000$. On bank marketing, with one split, it matches the real sample on single columns (Appendix E.5). Under the three paper protocols it is within 0.01 of CoRTeC on every utility measure (Appendix E.6).

Passing the decoder's own pool through Algorithm 2, as every CoRTeC arm does, closes the remaining gap, and the result holds on all three splits. The decoder with selection is below the plain decoder on every fidelity measure (0 of 3 seeds against, $p \leq 0.03$ except the held-out conditionals at $n = 300$). It is below a real sample of the same size on the seen conditionals at both sizes and on single columns at $n = 300$. It is within 0.014 AUC of the reference AIM on every student: lower on the forest at both sizes and on the linear student at $n = 1{,}000$, and higher on gradient boosting at $n = 300$ (3 of 3, $p = 0.01$). The release the auto-configurator derives on Adult, decoded the same way, improves further. Over the three splits its single columns are closer to the private table than the reference AIM's at both sizes (0 of 3 seeds in AIM's favour, $p \leq 0.01$), and so are its pairs at $n = 300$ ($p = 0.05$). Its students reach AIM's: within 0.007 AUC at $n = 1{,}000$ ($p \geq 0.35$ on all three), and higher on gradient boosting at $n = 300$ (3 of 3, $p = 0.03$). AIM fits pairs more closely at $n = 1{,}000$ (3 of 3, $p < 0.01$). Its conditional cells are not compared, because the evaluator's seen bands are the fixed rule's cohorts and two of the held-out families are columns it conditions on. On the one clean family it is closer to the private table than AIM at both sizes (Appendix E.2).

5.3 What each part contributes

Most of the gain comes from the selection step, not from the model. The auto-configured release improves on the plain decoder's single columns by 0.038 at $n = 300$ and 0.025 at $n = 1{,}000$. The selection step accounts for 0.023 and 0.014 of that, and the auto-configured workload for 0.015 and 0.011, each at 0 of 3 seeds against ($p \leq 0.03$). On the students each step adds 0.003 to 0.009, which is significant only for the selection step on the forest at $n = 1{,}000$ ($p = 0.01$).

Compared with the plain decoder, the model's overall effect is neutral on Adult, and negative on NHANES pairs at every seed ($p \leq 0.02$ at both sizes) and on bank at both sizes (Appendix E.5). Its one consistent gain, on the seen conditionals on Adult at $n = 300$ (3 of 3 seeds, $p = 0.04$), comes from the selection step and not from the model. The decoder's own pool passed through Algorithm 2 is better than both (0 of 3 seeds in the plain decoder's favour, $p \leq 0.01$). Pair by pair, the model's error is below the decoder's on pairs it recognises as logically linked, such as education with education number and marital status with relationship on Adult, or days since contact with the previous outcome on bank. Its error is above the decoder's on pairs it believes are related when the data says they are not, such as education with contact and loan with contact on bank (Appendix E.3). The model repairs what it knows and invents what it expects. Without the release, the same Gemini backbone writing rows from the column names alone is worse than the shuffled floor on the seen conditionals. About 70 % of its rows are verbatim copies of the first 25 rows of the public file (Appendix F.3), yet its students stay within 0.025 AUC of every CoRTeC arm. On a public benchmark, then, the fidelity columns separate the arms and the utility columns barely do. This is a sharper form of Swanberg et al.'s finding [37]. With the release, the same backbone copies none of those rows.

Take away the column names and the model's advantage goes with them. The anonymised schema renames every column, category and label to a neutral token. The release and the prompt are mapped in the same way, and the draws are mapped back before scoring, so the model sees the same numbers with no meaning attached (Appendix C). Table 4 gives Adult at $n = 300$. On Gemini 3.6 Flash the pair error rises to the decoder's level and the students fall below the decoder's. Only the single-column advantage survives, and it belongs to the selection step. The single-column error stays at 0.030, against the decoder's 0.058, because the selection step matches the pool to the released marginals whatever the names. On Qwen3-14B every measure is worse without names, and the same holds for Qwen3-32B over two draws. This matches DP-LLMTGen's finding that meaningful names help a fine-tuned model [38].

Table 4: The anonymised-schema control on Adult at ε = 2 and n = 300, using the same releases under neutral names. The 32B arm has two draws and the others three.

backbone	names	1-way TV ↓	2-way TV ↓	cond. seen ↓	cond. held-out ↓	TSTR-LR ↑	TSTR-RF ↑	TSTR-GBM ↑
Gemini 3.6 Flash	real	0.027	0.110	0.013	0.042	0.805	0.863	0.838
Gemini 3.6 Flash	neutral	0.030	0.148	0.014	0.044	0.745	0.840	0.798
Qwen3-14B	real	0.046	0.139	0.027	0.048	0.807	0.862	0.833
Qwen3-14B	neutral	0.065	0.168	0.044	0.054	0.811	0.846	0.817
Qwen3-32B (AWQ)	real	0.049	0.138	0.031	0.046	0.815	0.864	0.840
Qwen3-32B (AWQ)	neutral, 2 draws	0.060	0.159	0.067	0.052	0.785	0.858	0.822
maximum-entropy decoder	*none*	*0.058*	*0.146*	*0.082*	*0.049*	*0.818*	*0.856*	*0.847*
real sample		*0.043*	*0.107*	*0.062*	*0.036*	*0.830*	*0.870*	*0.840*

5.4 The release trades pairs for the conditionals it names

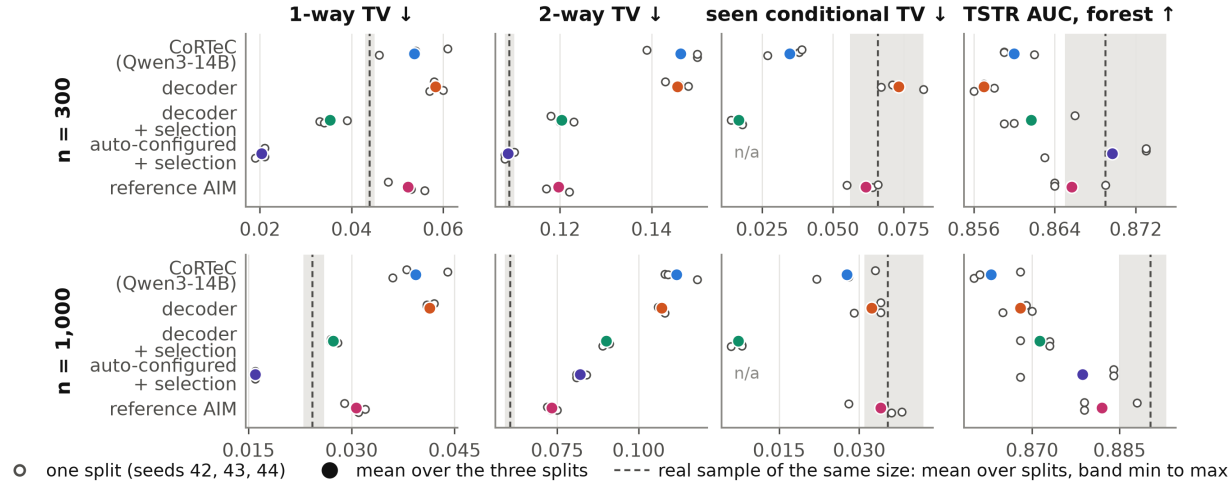


Figure 3: Replication on Adult over seeds 42, 43 and 44, each with a fresh split and a fresh release. An open circle is one split's mean over three draws, and a filled dot is the mean over the three splits. The dashed line and the grey band show a real sample of the same size: its mean, and its range over the splits. Four of the seven measures are shown at both sizes. The auto-configured release's conditional cells are not comparable (n/a). Appendix E.2 gives every seed, every measure, the standard deviations across seeds and every paired test.

The results replicate, and the cost of the design shows up on pairs of columns. Over three seeds and three datasets, the release reaches the utility of the reference AIM in five of six dataset-size settings. AIM's adaptive workload fits pairs more closely at n = 1,000 on Adult and at both sizes on NHANES, and the release carries the conditional rates it names on Adult more closely. We say the release reaches AIM's utility on a student when its mean is higher, or within 0.01 AUC with no consistent sign across the seeds. The one setting where it does not is NHANES at n = 300.

On Adult, the reference AIM's pair error is below CoRTeC's at every seed and at both sizes (p = 0.03 and below 0.01). AIM's students score higher than CoRTeC's at n = 1,000 at every seed (p = 0.07 to 0.08), and at n = 300 only the random forest separates them (p = 0.02). With the auto-configured release the utility gap closes at both sizes (p = 0.28 to 0.86). Its gradient-boosting student scores higher than AIM's at n = 300 at 3 of 3 seeds (p = 0.03), and AIM fits pairs more closely at n = 1,000 only. The difference on single columns comes from cohort size. The fixed rule forms twelve cohorts of 380 to 6,041 records, and the auto-configurator three cohorts of 4,434 to 13,345. A Laplace histogram has the same absolute noise whatever the count, so larger cohorts give smaller relative errors, while smaller cohorts give the named conditional rates. The conditional claim therefore belongs to the fixed rule, and the utility and single-column claims to the auto-configurator.

On NHANES, AIM fits pairs more closely than CoRTeC at every seed. Its students score higher at n = 300 (0 of 3 seeds in CoRTeC's favour, p = 0.12 to 0.18), and the two are level at n = 1,000 (p = 0.73 to 0.96). On single columns they are level at n = 300, and AIM is closer by 0.009 at n = 1,000. On Adult, CoRTeC's seen conditionals are closer to the private table than AIM's at both sizes ($p < 0.01$ at n = 300). Appendix E.1 gives the DP fine-tuning rows, DP-2Stage and Tab-PE at the same budget, on every measure.

5.5 The privacy architecture and the cost of a draw

No private row reaches the model, and nothing is copied into a draw. Five backbones decoded the same Adult release with the same prompts and settings (Appendix E.4). Single-column fidelity improves with model size and levels off at 14 billion parameters, while utility does not change with the backbone. The same Qwen3-14B fine-tuned under DP-SGD at the same budget is further from the private table on both fidelity measures at both sizes (Table 3), so where the budget is spent matters more than which backbone is used. Membership inference (Appendix F) runs four attacks [11, 33, 34] on 3,000 candidates, half of them training rows and half held out, and each attack is validated on a positive control. The strongest advantage is at most 0.035 on every Adult arm and 0.026 to 0.060 on NHANES, against a permitted 0.762. The distance-to-closest-record test finds no synthetic row that copies a training row, and no arm that is closer to the training rows than to a held-out set (Appendix F). The served Gemini backbone can recite rows 2 to 25 of the public Adult file, but none of those rows appears in any CoRTeC draw (Appendix F.3). A 300-row Gemini draw costs 1.56 dollars at list price and the same draw on Qwen3-14B about 50 minutes on one rented A40, while the release takes seconds and the maximum-entropy decoder runs on a CPU. AIM fits once and then samples for free, which suits large draws (Appendix G gives an indicative summary in percent).

6. Limitations

The release spends the budget on the conditional rates a deployment names, at a cost on pairs of columns that the reference AIM does not pay. AIM fits pairs more closely at n = 1,000 on Adult and at both sizes on NHANES. Its students score higher on NHANES at n = 300, and higher than the fixed-rule release's on Adult at n = 1,000. It also fits the seen conditionals more closely on NHANES at n = 300 and on bank marketing. Our draws are at most 1,000 rows, so utility at larger release sizes is not measured. The gap to a real sample of the same size is largest on the medical dataset, 0.07 AUC at n = 300. The language model's contribution depends on the column names, so an institution whose schema the model does not know should expect the decoder's result rather than the Gemini arm's. Several results come from a single split: the backbone ladder, the anonymised and one-shot controls, the decoder at ε = 0.5 and 1, the bank dataset and the paper-protocol rows. The 8B arm and the anonymised 32B arm have two draws, and the Gemini arms ran at n = 300 only. The auto-configured release was decoded with no model only, so what a model adds to it is not measured, and the Qwen3 backbones were not given the memorisation test. The reference AIM runs on CoRTeC's public bins, a conservative version of the PrivTree-binned AIM that Tab-PE's authors run [39]. We make no claim about high-order correlations, the axis on which Tab-PE's stress test separates methods [39]. Our rerun of the DP-LLMTGen procedure understates that method (Appendix E.6), so its published row is the one to use as the reference. Every real dataset here is a public benchmark, and the anonymised control removes what the model knows about the schema but not what it may know about the marginals. CoRTeC draws whose pool fell short of the requested size are scored as delivered (Appendix D), which costs them sampling error that the references do not pay.

7. Conclusion

For a regulated institution, the product is the release itself: class-conditional histograms and a conditional table inside cohorts, with the workload fixed before the release is computed. The budget is spent once, on a file of counts that anyone can inspect and decode at no further cost. Decoded without a language model, it trains models close to a real sample of its size and reaches the utility of the reference AIM in five of six settings, at a cost on pairs of columns. A frozen language model decoding the same file adds what it knows from the column names. It repairs the pairs it recognises, invents the pairs it expects, and owes its advantage on single columns to the selection step. The main argument for the design is its privacy architecture. The model never sees a row, every draw is post-processing, and the backbone can change without touching the guarantee. The measured membership advantage of the CoRTeC draws, 0.02 to 0.05, is a small fraction of the 0.762 the budget permits. CoRTeC is meant to sit beside the marginal methods,

not to replace them: AIM suits a deployment that needs pairs of columns, and CoRTeC one that needs named conditional rates, a file it can inspect and decoding it can repeat. For a deployment, we recommend decoding with the maximum-entropy model and the selection step first, and adding a language model only where it knows the schema.

Ethics Statement

This work uses public benchmark datasets and a national health survey released for research. No private institutional data was used. The membership attacks and memorisation tests were run on synthetic data we generated, and against a served commercial model with the public Adult file, which that model can partly reproduce. The method is meant to reduce the privacy risk of sharing tabular data. Its guarantee covers the release and its draws, not the model's pretraining corpus, and its limits are stated where they arise.

Reproducibility Statement

The repository (Calyie/cortec) holds the Apache-2.0 reference implementation with its tests, the research pipeline that produced every CoRTeC arm, the paper and the frozen auto-configurations. Released statistics cannot be regenerated bit for bit, because the noise comes from a cryptographically secure source that ignores any seed. Every stored release is therefore kept, and every draw reuses it by file. The reference-AIM wrapper, the maximum-entropy decoder, the decoder-with-selection and one-shot controls, the anonymisation of releases and the scorers are in the repository. The released implementation decodes a release with no model as `cortec decode` (Private-PGM and the selection step). The synthetic draws, the per-draw score files, the membership-attack files and the stored releases of every arm are deposited with the Zenodo record of this paper (10.5281/zenodo.22945315). Every number in this paper can be traced to a stored result file.

AI-Use Statement

Language models appear in this work as the objects of study: the generators named in every table. Language-model assistants were also used to draft and edit the text and the code. Every experiment was designed, run and checked by the author, and every number was produced by the described pipeline from stored result files.

Compute and Cost

The Qwen3 backbones and the DP fine-tuning baselines ran on one rented NVIDIA A40, with vLLM [45] for generation and Opacus [41] for DP-SGD. The reference AIM and MST fits and the maximum-entropy decoder ran on a CPU, and the Gemini arms ran through Google's interface. The named and anonymised Gemini 3.6 Flash pairs cost 9.37 dollars in total, the memorisation test 0.33 dollars, and the one-shot control 2.52 dollars (156 calls). A 300-row CoRTeC draw takes 41 to 46 calls of 25 rows.

References

[1] Martín Abadi, Andy Chu, Ian Goodfellow, H. Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. Deep learning with differential privacy. In *ACM SIGSAC Conference on Computer and Communications Security (CCS)*, pp. 308–318, 2016.

[2] Meenatchi Sundaram Muthu Selva Annamalai and Emiliano De Cristofaro. Nearly tight black-box auditing of differentially private machine learning. In *NeurIPS*, 2024.

[3] Tejumade Afonja, Hui-Po Wang, Raouf Kerkouche, and Mario Fritz. DP-2Stage: Adapting language models as differentially private tabular data generators. *Transactions on Machine Learning Research*, 2025.

[4] Sebastian Bordt, Harsha Nori, and Rich Caruana. Elephants never forget: Testing language models for memorization of tabular data. In *NeurIPS Table Representation Learning Workshop*, 2023.

[5] Vadim Borisov, Kathrin Seßler, Tobias Leemann, Martin Pawelczyk, and Gjergji Kasneci. Language models are realistic tabular data generators. In *International Conference on Learning Representations (ICLR)*, 2023.

[6] Claire McKay Bowen and Joshua Snoke. Comparative study of differentially private synthetic data algorithms from the NIST PSCR differential privacy synthetic data challenge. *Journal of Privacy and Confidentiality*, 11(1), 2021.

[7] Kuntai Cai, Xiaoyu Lei, Jianxin Wei, and Xiaokui Xiao. Data synthesis via differentially private Markov random fields. *Proceedings of the VLDB Endowment*, 14(11):2190–2202, 2021.

[8] Rodrigo Castellon, Achintya Gopal, Brian Bloniarz, and David Rosenberg. DP-TBART: A transformer-based autoregressive model for differentially private tabular data generation. *arXiv:2307.10430*, 2023.

[9] CDC/NCHS. National Health and Nutrition Examination Survey (NHANES) 2017–2018 data files. National Center for Health Statistics, 2020.

[10] Kai Chen, Xiaochen Li, Chen Gong, Ryan McKenna, and Tianhao Wang. Benchmarking differentially private tabular data synthesis. *Proceedings of the ACM on Management of Data*, 3(6), 2025.

[11] Nicholas Carlini, Steve Chien, Milad Nasr, Shuang Song, Andreas Terzis, and Florian Tramèr. Membership inference attacks from first principles. In *IEEE Symposium on Security and Privacy*, pp. 1897–1914, 2022.

[12] W. Edwards Deming and Frederick F. Stephan. On a least squares adjustment of a sampled frequency table when the expected marginal totals are known. *Annals of Mathematical Statistics*, 11(4):427–444, 1940.

[13] Dheeru Dua and Casey Graff. UCI machine learning repository. University of California, Irvine, 2019.

[14] Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. Calibrating noise to sensitivity in private data analysis. In *Theory of Cryptography Conference (TCC)*, pp. 265–284, 2006.

[15] Cynthia Dwork and Aaron Roth. The algorithmic foundations of differential privacy. *Foundations and Trends in Theoretical Computer Science*, 9(3–4):211–407, 2014.

[16] Georgi Ganev, Meenatchi Sundaram Muthu Selva Annamalai, and Bogdan Kulynych. Tight auditing of differential privacy in MST and AIM. In *Theory and Practice of Differential Privacy (TPDP)*, 2026.

[17] Georgi Ganev, Bristena Oprisanu, and Emiliano De Cristofaro. Robin Hood and Matthew effects: Differential privacy has disparate impact on synthetic data. In *ICML*, pp. 6944–6959, 2022.

[18] Michael Hay, Ashwin Machanavajjhala, Gerome Miklau, Yan Chen, and Dan Zhang. Principled evaluation of differentially private algorithms using DPBench. In *ACM SIGMOD*, pp. 139–154, 2016.

[19] Jingang Qu, David Holzmüller, Gaël Varoquaux, and Marine Le Morvan. TabICL: A tabular foundation model for in-context learning on large data. In *International Conference on Machine Learning (ICML)*, 2025.

[20] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.

[21] Florimond Houssiau, James Jordon, Samuel N. Cohen, Owen Daniel, Andrew Elliott, James Geddes, Callum Mole, Camila Rangel-Smith, and Lukasz Szpruch. TAPAS: A toolbox for adversarial privacy auditing of synthetic data. In *NeurIPS Workshop on Synthetic Data for Empowering ML Research*, 2022.

[22] James Jordon, Jinsung Yoon, and Mihaela van der Schaar. PATE-GAN: Generating synthetic data with differential privacy guarantees. In *International Conference on Learning Representations (ICLR)*, 2019.

[23] Ron Kohavi and Barry Becker. Adult data set. UCI Machine Learning Repository, 1996.

[24] Zinan Lin, Sivakanth Gopi, Janardhan Kulkarni, Harsha Nori, and Sergey Yekhanin. Differentially private synthetic data via foundation model APIs 1: Images. In *International Conference on Learning Representations (ICLR)*, 2024.

[25] Sérgio Moro, Paulo Rita, and Paulo Cortez. Bank marketing data set. UCI Machine Learning Repository, 2014.

[26] Ryan McKenna, Daniel Sheldon, and Gerome Miklau. Graphical-model based estimation and inference for differential privacy. In *International Conference on Machine Learning (ICML)*, pp. 4435–4444, 2019.

[27] Ryan McKenna, Gerome Miklau, and Daniel Sheldon. Winning the NIST contest: A scalable and general approach to differentially private synthetic data. *Journal of Privacy and Confidentiality*, 11(3), 2021.

[28] Ryan McKenna, Brett Mullins, Daniel Sheldon, and Gerome Miklau. AIM: An adaptive and iterative mechanism for differentially private synthetic data. *Proceedings of the VLDB Endowment*, 15(11):2599–2612, 2022.

[29] Frank McSherry. Privacy integrated queries: An extensible platform for privacy-preserving data analysis. In *ACM SIGMOD*, pp. 19–30, 2009.

[30] Nicolas Papernot, Martín Abadi, Úlfar Erlingsson, Ian Goodfellow, and Kunal Talwar. Semi-supervised knowledge transfer for deep learning from private training data. In *International Conference on Learning Representations (ICLR)*, 2017.

[31] Nicolas Papernot, Shuang Song, Ilya Mironov, Ananth Raghunathan, Kunal Talwar, and Úlfar Erlingsson. Scalable private learning with PATE. In *International Conference on Learning Representations (ICLR)*, 2018.

[32] Lucas Rosenblatt, Xiaoyan Liu, Samira Pouyanfar, Eduardo de Leon, Anuj Desai, and Joshua Allen. Differentially private synthetic data: Applied evaluations and enhancements. *arXiv:2011.05537*, 2020.

[33] Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. Membership inference attacks against machine learning models. In *IEEE Symposium on Security and Privacy*, pp. 3–18, 2017.

[34] Theresa Stadler, Bristena Oprisanu, and Carmela Troncoso. Synthetic data: Anonymisation groundhog day. In *USENIX Security Symposium*, pp. 1451–1468, 2022.

[35] Ilya Mironov. On significance of the least significant bits for differential privacy. In *ACM CCS*, pp. 650–661, 2012.

[36] Yuchao Tao, Ryan McKenna, Michael Hay, Ashwin Machanavajjhala, and Gerome Miklau. Benchmarking differentially private synthetic data generation algorithms. *arXiv:2112.09238*, 2021.

[37] Marika Swanberg, Ryan McKenna, Edo Roth, Albert Cheu, and Peter Kairouz. Is API access to LLMs useful for generating private synthetic tabular data? *arXiv:2502.06555*, 2025.

[38] Toan V. Tran and Li Xiong. Differentially private tabular data synthesis using large language models. *arXiv:2406.01457*, 2024.

[39] Toan Tran, Arturs Backurs, Zinan Lin, Victor Reis, Li Xiong, and Sergey Yekhanin. Differentially private synthetic data via APIs 4: Tabular data. In *International Conference on Machine Learning (ICML)*, 2026.

[40] Florian Tramèr, Gautam Kamath, and Nicholas Carlini. Position: Considerations for differentially private learning with large-scale public pretraining. In *International Conference on Machine Learning (ICML)*, 2024.

[41] Ashkan Yousefpour, Igor Shilov, Alexandre Sablayrolles, Davide Testuggine, Karthik Prasad, Mani Malek, John Nguyen, Sayan Ghosh, Akash Bharadwaj, Jessica Zhao, Graham Cormode, and Ilya Mironov. Opacus: User-friendly differential privacy library in PyTorch. *arXiv:2109.12298*, 2021.

[42] Lei Xu, Maria Skoularidou, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. Modeling tabular data using conditional GAN. In *NeurIPS*, 2019.

[43] Chulin Xie, Zinan Lin, Arturs Backurs, Sivakanth Gopi, Da Yu, Huseyin A. Inan, Harsha Nori, Haotian Jiang, Huishuai Zhang, Yin Tat Lee, Bo Li, and Sergey Yekhanin. Differentially private synthetic data via foundation model APIs 2: Text. In *International Conference on Machine Learning (ICML)*, 2024.

[44] Liyang Xie, Kaixiang Lin, Shu Wang, Fei Wang, and Jiayu Zhou. Differentially private generative adversarial network. *arXiv:1802.06739*, 2018.

[45] Woodrow Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *SOSP*, 2023.

[46] Jun Zhang, Graham Cormode, Cecilia M. Procopiuc, Divesh Srivastava, and Xiaokui Xiao. PrivBayes: Private data release via Bayesian networks. *ACM Transactions on Database Systems*, 42(4), 2017.

[47] Chaoyi Zhu, Jiayi Tang, Juan F. Pérez, Marten van Dijk, and Lydia Y. Chen. DP-TLDM: Differentially private tabular latent diffusion model. In *International Conference on Availability, Reliability and Security (ARES)*, 2025.

Appendix A. Privacy Analysis

Proof of Theorem 1. Let D and D' be neighbours that differ in one record x.

(i) *Forming the cohorts is free.* Π is a function of public attribute domains, so $\Pi(x)$ is computed from x's own attributes without looking at any other record. The partition does not depend on the data, and releasing it costs nothing.

(ii) *Within a cohort, sequential composition.* Fix a cohort P. Each released histogram is a counting query of L1 sensitivity 1, because x contributes to exactly one bin of one attribute. Removing x changes that histogram's L1 norm by 1 and leaves every other histogram unchanged. The Laplace mechanism at scale $1/\varepsilon_q$ makes each query ε_q-differentially private. Under the pooled block, the q queries on P compose sequentially to $q\ \varepsilon_q = \varepsilon_{\text{marg}}$. Under the class-conditional blocks, P is split by y, so x lies in exactly one of $P \cap \{y = 0\}$ and $P \cap \{y = 1\}$. It is touched by the class-balance query and by the $|A|$ histograms of its own block, and by nothing in the other block. That is again a chain of q queries costing $q\ \varepsilon_q$. The two blocks do not overlap and compose in parallel. The pooled histogram is computed from released quantities only.

(iii) *Across cohorts, parallel composition.* The cohorts do not overlap. They are defined by a public rule or, under auto-configuration, by the released output of the charged selection, which Theorem 1 charges as ε_{sel}. The record x lies in exactly one cohort, so it can affect only that cohort's queries. The cost across all cohorts is therefore $\varepsilon_{\text{marg}}$, however many cohorts there are.

(iv) *Within a conditional level, parallel composition.* The cells of a level partition D, so the level costs ε_L however many cells it contains. Within a cell, the positive count and the cell size each have sensitivity 1 under add/remove adjacency. Both are noised with $\text{Lap}(2/\varepsilon_L)$, a scale that depends on nothing private, and since they touch the same records they compose sequentially to ε_L. The published rate $\tilde{k_c} / \max(\tilde{n_c}, n_{\min})$, clipped to [0, 1], is post-processing of two released counts and the public floor. We do not noise the rate directly with a scale set by the private size $|c|$. Two Laplace densities with different scales have an unbounded likelihood ratio in one tail, so that mechanism would not be pure ε.

(v) *Across levels, sequential composition, and the cohort sizes.* The levels describe the same people and are not disjoint, so their budgets add: $|L|\ \varepsilon_L = \varepsilon_{\text{cond}}$. Each cohort size is a counting query of sensitivity 1 noised with $\text{Lap}(1/\varepsilon_{\text{count}})$. The cohorts do not overlap, so all the sizes together cost $\varepsilon_{\text{count}}$, which composes sequentially with each cohort's other queries. Every $n_{\min}$ decision compares a noised count with the public floor, and every cohort and cell of the public domains is queried whether or not it holds a record. Which groups appear in R is therefore post-processing.

Stage B is post-processing. The row allocation reads released sizes. The per-bin counts a batch is asked for are apportioned from released histograms, and the counts still owed are the difference between released targets and generated rows. The prompt carries R and those counts, and the frozen model reads only the prompt. The acceptance rule reads public domains. The inclusion weights, the raking, the refinement and the sub-bin redraw read R, the generated rows and fresh randomness. None of these steps increases the privacy loss, at any pool factor and for any number of draws. The maximum-entropy decoder reads only R and is post-processing in the same way.

Adding the parts gives $\varepsilon_{\text{count}} + \varepsilon_{\text{marg}} + \varepsilon_{\text{cond}} = \varepsilon_{\text{total}}$. Under auto-configuration, the noised contingency tables of the candidate columns are counting queries of sensitivity 1 on the same records, and they compose sequentially to ε_{sel}. Everything computed from their noised counts is post-processing. The release that follows spends $\varepsilon_{\text{release}}$, and $\varepsilon_{\text{sel}} + \varepsilon_{\text{release}} = \varepsilon_{\text{total}}$. Stage C's k rate queries are taken over $\max(|c|, n_{\min})$, so each has sensitivity at most $1/n_{\min}$. They act on disjoint cells with noise of public scale, and together they cost $\varepsilon_{\text{cert}}$. ∎

The floating-point attack on the Laplace mechanism [35] applies to every implementation in the comparison, including ours, and the release's audit block lists it as an open gap. The empirical privacy of the private-pgm mechanisms we compare against has been audited and found close to their stated guarantee [16].

Appendix B. Metrics, Floors and Tests

Fidelity. The 1-way total variation of a column is half the summed absolute difference between its synthetic and private histograms over the public bins. We average it over columns. The 2-way total variation is the same quantity over the joint histogram of each pair of columns, averaged over pairs. Both are measured against the full private training table. Conditional total variation is the mean absolute difference between $P(y = 1 \mid g)$ in the synthetic table and in the private table, over a family of groups g. We use two families: the cohort bands of the stratification rule ("seen", the groups every release describes), and groups built from columns that share nothing with the conditioning ("held-out").

Utility. We measure train-on-synthetic, test-on-real AUC with three students: logistic regression (balanced, 2,000 iterations, standardised inputs), a random forest (300 trees, balanced) and gradient boosting (scikit-learn's histogram implementation). Each is trained on the synthetic table and scored on the real test split of the same seed. We report the three students separately, because a result can change sign between a linear and a tree student.

References in every table. The real sample is a fresh draw of n rows from the training split, drawn once per table at that table's n, so its value varies between tables by design. The permuted target is the same rows with the target column shuffled, which keeps every marginal and removes every link between the features and the target. The ceiling trains the students on a real sample of 4,000 training rows, or on the whole training split where it holds fewer (NHANES, 2,999 rows). Its fidelity cells are that sample's distance from the full training table: the sampling error at 4,000 rows, and 0 where the sample is the whole split. The two floors separate on every conditional and utility measure but not on the marginal ones, which is why no marginal measure is read on its own.

Repetition and tests. Every cell is the mean over three draws from one release, with the sample standard deviation. The replication draws a fresh split and a fresh release at seeds 43 and 44. For each pair of conditions it reports the mean paired difference over the three seeds, the sign count and a two-sided paired t-test (scipy). With three pairs the test has little power, so a difference is read as replicated when its sign is the same at every seed, and its p-value is reported beside it.

The bin convention. The evaluator's intervals are closed on the right and the release's on the left, so an integer that falls exactly on an edge is counted in different bins by the two. On Adult this changes 1-way error by 0.003 and changes no ordering.

Appendix C. Datasets, Preprocessing and the Anonymised Schema

Adult. We use the 32,561 rows of the UCI training file and keep "?" as a category. There are 6 numerical columns (age, final weight, education number, capital gain, capital loss, hours per week) and 8 categorical ones, and the target is income over 50K. The public bins follow the edges declared in the schema. The cohorts come from the schema's own rule over education and hours per week, and the conditional levels are the first and the fourth of the declared hierarchy. The auto-configured Adult release (Section 5, Appendix D) takes its cohorts and levels from the auto-configurator instead.

NHANES 2017-2018. We join the demographic, glycohaemoglobin, body-measures and blood-pressure files on the respondent identifier and keep adults aged 18 to 80. There are 5 numerical columns (age, BMI, waist, systolic and diastolic blood pressure) and 4 categorical ones (sex, race and ethnicity, education, income bracket), and the target is HbA1c at or above 6.5 %. The bin edges are public clinical knowledge: WHO BMI categories and ACC/AHA blood-pressure stages. The release is auto-configured. The auto-configurator's charged column ranking chose three age cohorts and two conditional levels over age, diastolic blood pressure, and race and ethnicity.

Bank marketing. We use the UCI file of 45,211 rows and 13 of its 16 input columns. Five are numerical (age, balance, call duration, campaign contacts, days since the previous campaign) and eight categorical (job, marital status, education, default, housing loan, personal loan, contact type, previous outcome). The day, month and previous-contact-count columns are not used. The target is a term-deposit subscription, the split is 80/20 at seed 42, and the release is auto-configured in the same way as NHANES.

The anonymised schema. A fixed public rule replaces every column name, category label, band label and the dataset description with a neutral token. Numerical columns become n1 to n6 in schema order, categorical columns k1 to k8, and the target y with classes A and B. Each categorical column's sorted categories become k3_v01, k3_v02 and so on, and numerical band labels become n3_b1, n3_b2 and so on. Numeric values, public bounds, bins, the cohort rule, the conditional levels and the held-out families are unchanged. A release of Adult therefore maps one to one onto a release of the anonymised schema, and the model sees the same prompt templates with the meaning removed. The draws are mapped back before scoring, so the scorer is the same.

Appendix D. Configuration, the Prompt and the Baselines

CoRTeC settings. $\varepsilon = 2$ (0.5 and 1 where stated); conditional share $\alpha = 0.2$; count share $\gamma = 0.02$; $n_{\min} = 150$; 25 rows per call; counts stated per outcome class; pool factor 3 (2 on NHANES); selection with the sub-bin rule; one cohort worker; seed 42 for the generator and the selection. The Qwen3 backbones ran through vLLM [45] with each model's

default sampling, and the Gemini backbone ran with reasoning enabled through Google's interface. A CoRTeC draw is scored as delivered. Where the pool fell short of the requested size, the draw holds fewer than n rows: between 276 and 300 rows at n = 300 and between 949 and 1,000 at n = 1,000 on the Qwen3 arms, between 257 and 279 and between 807 and 864 on the Qwen2.5-7B arm, and within 1 % of n on the Gemini and bank marketing arms. The real-sample and permuted references in every table are drawn at the full n. Fewer rows add sampling error to a draw's own fidelity measures, so the shortfall works against CoRTeC and against no other method. A call whose output does not parse is retried, and the retry and parse rates for each backbone are in the stored manifests. Qwen3-14B produced no parse failure in about 1,500 calls. Qwen3-8B writes the prompt's range labels into numerical columns when a quota is small. The quantised 32B fails about three NHANES calls in four on one cohort and completes when resumed.

The prompt. The system message and the cohort template are fixed and fingerprinted in the reference implementation. The system message asks the model to reproduce the distributions and relationships the statistics describe "even if this contradicts your expectations about how these variables usually relate", to treat the statistics as describing this specific population, and to output only CSV rows. The cohort template states the subgroup, the released numerical histograms, the released categorical shares, the outcome rate, and the conditional outcome table, which it calls the most important part of the prompt. It then gives the exact counts the batch must carry for each bin, category and outcome, the output format, and four rules: values inside their ranges, the target as one of its two labels, no repeated rows, and nothing but the header and the rows. In a cohort with class blocks, the numerical and categorical blocks are given once for each outcome class. The full text is in the reference implementation's template file and in the deployment guide.

The integer rule. Whether a numerical column holds integers is decided from the public bin edges and bounds, which must all be whole numbers, together with the majority of the pool's values. A rule that looks at every value instead lets a single model-written "45.0" or "37.5" turn the column into real values. The sub-bin redraw then produces fractional ages, hours and capital values, which an evaluator comparing exact values treats as far from the data. The anonymised Qwen3-14B draws, selected under that rule during development, scored 0.117 on single columns before flooring and 0.065 after. Every CoRTeC draw in this paper is scored after its integer columns are floored, and the released implementation carries the majority rule with a regression test. The Tab-PE and DP-2Stage draws are scored as their methods deliver them, with fractional values in integer columns. Flooring those columns would lower their 1-way and 2-way distances by at most 0.011 and 0.017 on the draws we measured, so no ordering depends on it. Two anonymised Qwen3-32B draws have 25 and 24 empty hours-per-week cells, which the scorer drops from that column's histogram.

Reference AIM and MST. We use private-pgm (mbi 1.1) at the authors' defaults: an 80 MB maximum model size, a 2-way workload, 1,000 mirror-descent iterations, the exponential mechanism for AIM's selection, $\delta = 10^{-9}$, and the corrected sampler for both. During development we also ran the smartnoise-synth implementations of AIM and MST at their library defaults. They carry the sampler defect described below, their students score 0.15 to 0.17 AUC below the reference implementation's on Adult at n = 300, and no number from them appears in this paper. Every AIM and MST number here comes from the authors' private-pgm.

The wrapper is invoked as `pgm_baselines.py --train <split> --dataset <name> --epsilon 2 --n-out <n> --draws 3 --seed 42 --methods aim mst --max-model-size 80 --degree 2 --max-iters 1000`. For the reference AIM rows of Table 3 and of the replication, `PGM_ORACLE=stable` and `PGM_CONSISTENCY_GATE=0.08` are set in the environment, with three draws per seed under the stable oracle so that the three seeds are like for like. The reference MST rows and the $\varepsilon = 0.5$ and 1 rows of Appendix E.7 are five draws (`--draws 5`) from the wrapper's runs under the library's default oracle, which is why the caption of Table 3 counts five draws for MST. The wrapper is in the repository with the decoder, the scorers and the anonymisation map.

Under mbi's default message-passing oracle, the fit raised "probabilities contain NaN" inside mirror descent on the seed-43 and seed-44 splits of both datasets, and on seeds 45 to 47, 53, 54 and 60 to 63 in a sweep. The library's fallback, a halved step, returned near-uniform models. On seed 43, NHANES, n = 300, such a model gave a 1-way total variation of 0.182, students at 0.50 and a diabetes rate of 43 % against 14 %. None of those fits is reported. With the log-space oracle (`marginal_oracles.message_passing_stable`) the same seeds converge at the first attempt. On the seed-42 Adult split the stable refit matches the default-oracle fit to within 0.002 on single columns and pairs (0.029 and 0.072 against 0.031 and 0.074 at n = 1,000) and within 0.001 AUC on the students (0.854, 0.888 and 0.883 against 0.853, 0.887 and 0.883). The consistency check is the mean total variation between the fitted marginals and the noisy measurements they were fitted to, which reads released values only. The library's threshold is

0.15. We set it to 0.08, because the first-attempt fits on every seed and size measure between 0.005 and 0.036, so a threshold of about twice the largest accepted value rejects a fallback fit and accepts every legitimate one. Every reported fit passed at its first attempt.

The corrected sampler fixes a defect in mbi 1.1's `synthetic_data`. That function reads each child column's conditional table in the model's attribute order, not in the order in which the parents were sampled. Where the two orders differ, the joint structure of some columns is scrambled while every 1-way marginal survives. The wrapper samples each column from its conditional with the axes aligned, and the sampled conditionals then match the fitted model's to within 0.01.

The maximum-entropy decoder. Private-PGM's estimator is given exactly the released cohort sizes, class balances, histograms and conditional table of CoRTeC's release as its measurements, with no further noise and no further query, and it is sampled at the requested n.

The decoder with the selection step. On each of the three splits, the decoder of that split's release is sampled at three times n, giving three pools per size. Each pool is passed through Algorithm 2 with the sub-bin rule, using the seeds the CoRTeC arms use, and the integer columns are floored. The selection fell one to three rows short of n in six of the eighteen draws. This is post-processing of the release, with no privacy cost, and it runs on a CPU.

The auto-configured Adult release. On each split we ran the released implementation's Stage A (`release_statistics` with the auto-configurator). On the seed-42 and seed-44 splits it chose three age cohorts and a hierarchy of relationship, then relationship by marital status. On the seed-43 split it chose education-number cohorts and a hierarchy of relationship, then relationship by native country. The budget went 0.05 to the hierarchy selection, 0.05 plus the gate charge to the published counts, a fifth of the remainder to the conditional cells and the rest to the marginals. Each split used 105 queries, accounted $\varepsilon = 2.000$ and sealed the ledger. The release was decoded by `cortec decode` (the maximum-entropy fit, a pool of three times n and the selection step) with seeds 42 to 44, and the fit's consistency with the release was 0.004. No language-model arm was run on it.

The one-shot control. This uses the research pipeline's header-only prompt: the dataset's one-line description, the column names in order, the two labels of the target and a request for 25 realistic rows per call. It uses no release and no private data. The model is Gemini 3.6 Flash with reasoning at its default, as in the Gemini arms, with twelve calls per 300-row draw and forty per 1,000-row draw. Integer columns are floored and the draws are scored as delivered. The draws' positive rate is 36 to 38 %, against the data's 24 %, and about 70 % of their rows are verbatim copies of the public file's first 25 rows (Appendix F.3).

DP-2Stage. We ran the authors' released code: GPT-2, full fine-tuning, DP-Adam, clipping 1, $\delta = 10^{-5}$, loss weight 0.65, five pseudo-data epochs and ten private epochs, and temperature 0.7 with rejection sampling. It ran on our seed-42 split at $\varepsilon = 2$ and, under the authors' protocol, on their own split at $\varepsilon = 1$.

DP fine-tuning of the same backbones. We applied DP-LLMTGen's procedure to Qwen3-14B, Qwen2.5-7B and Llama-2 7B. Stage 1 trains for five epochs on random tables drawn from the public names, categories and ranges, at a learning rate of 10^{-4}. Stage 2 trains for two epochs of DP-SGD (Opacus [41]) on the private rows, with LoRA [20] of rank 8 on the attention projections (10.5 million trainable parameters on the 14B), a learning rate of 5×10^{-4}, batch size 32, clipping 1, the weighted cross-entropy with $\alpha = 0.65$, and $\delta = 10^{-5}$. It spent $\varepsilon = 1.994$ of 2. Sampling at temperature 0.7 gave valid rows on 99.7 % of attempts. We omit the original's numerical-understanding loss, because its coefficients are not given.

Tab-PE. We used the authors' DPSDA code and their Adult example: a mutation rate of 0.5 decaying polynomially to 0.01 with $\gamma = 0.2$, the composite population with sampling then ranking, the L2 nearest-neighbour histogram, $\delta = 1/(n \ln n)$, 29 to 30 iterations, and a noise multiplier of 11.3 at $\varepsilon = 2$ on our split. Populations were 1,000 rows (600 on NHANES) and, under the authors' protocol, 2,000 rows on their 70/15/15 split with 30 iterations.

Appendix E. Full Tables

E.1 NHANES at $\varepsilon = 2$, seed 42, three draws. Lower is better in the first four columns and higher in the last three.

condition, $n = 300$	1-way TV	2-way TV	cond. seen	cond. held-out	TSTR-LR	TSTR-RF	TSTR-GBM
all 2,999 training rows (utility ceiling)	*0.000*	*0.000*	*0.000*	*0.000*	*0.797*	*0.763*	*0.747*
real sample (floor)	*0.034*	*0.081*	*0.015*	*0.036*	*0.773*	*0.728*	*0.716*

condition, n = 300	1-way TV	2-way TV	cond. seen	cond. held-out	TSTR-LR	TSTR-RF	TSTR-GBM
permuted target	*0.043*	*0.093*	*0.047*	*0.060*	*0.493*	*0.514*	*0.515*
CoRTeC, Qwen3-14B	0.029	0.158	0.036	0.075	0.678	0.657	0.645
CoRTeC, Qwen3-32B (AWQ)	0.047	0.157	0.036	0.046	0.681	0.677	0.657
CoRTeC, Qwen3-8B	0.042	0.181	0.017	0.050	0.651	0.646	0.655
maximum-entropy decoder	0.042	0.117	0.025	0.041	0.705	0.694	0.691
reference AIM	0.042	0.100	0.017	0.044	0.747	0.721	0.710
reference MST (5 draws)	0.045	0.110	0.026	0.060	0.709	0.668	0.642
DP-2Stage, GPT-2	0.175	0.295	0.117	0.115	0.527	0.537	0.553
DP fine-tuned Qwen3-14B	0.116	0.205	0.074	0.077	0.575	0.544	0.527
Tab-PE (600-row population)	0.064	0.151	0.030	0.053	0.739	0.729	0.724
condition, n = 1,000							
real sample (floor)	*0.016*	*0.039*	*0.008*	*0.016*	*0.783*	*0.752*	*0.744*
permuted target	*0.017*	*0.044*	*0.039*	*0.047*	*0.516*	*0.484*	*0.470*
CoRTeC, Qwen3-14B	0.027	0.150	0.031	0.069	0.696	0.648	0.639
CoRTeC, Qwen3-32B (AWQ)	0.036	0.123	0.023	0.031	0.748	0.727	0.701
CoRTeC, Qwen3-8B	0.034	0.156	0.023	0.040	0.748	0.688	0.671
maximum-entropy decoder	0.031	0.088	0.022	0.029	0.758	0.720	0.686
reference AIM	0.029	0.070	0.027	0.032	0.752	0.714	0.660
reference MST (5 draws)	0.028	0.075	0.016	0.048	0.727	0.692	0.679
DP-2Stage, GPT-2	0.169	0.284	0.120	0.110	0.538	0.520	0.512
DP fine-tuned Qwen3-14B	0.117	0.195	0.089	0.078	0.571	0.516	0.501
Tab-PE (1,000-row population)	0.060	0.132	0.027	0.041	0.743	0.721	0.716

E.2 Replication: means over seeds, with the standard deviation across seeds, and the paired differences. The real-sample row is each split's own sample of n training rows, averaged over the three splits; it is the band in Figure 3. Each seed's value is the mean of three draws. p is a two-sided paired t-test over the three seeds, and the sign count gives how many seeds have a positive and a negative difference. The decoder-with-selection rows are the fixed-rule release decoded with no model and passed through the selection step on each split. The auto-configured rows are the release the auto-configurator derives on Adult, decoded the same way (Section 5.1). Their conditional cells are not comparable and are left out. The evaluator's seen bands are the fixed rule's cohorts, which that release does not condition on, and two of the three held-out families (relationship and native country) are columns it does condition on. On the one family that shares no column with its conditioning, workclass by race, its conditional error is 0.043 against the reference AIM's 0.061 at n = 300 and 0.020 against 0.036 at n = 1,000, with 0 of 3 seeds in AIM's favour at both sizes. The decoder reads 0.061 and 0.043.

Adult, n = 300	1-way TV	2-way TV	cond. seen	cond. held-out	TSTR-LR	TSTR-RF	TSTR-GBM
real sample (floor)	*0.044 ± 0.001*	*0.109 ± 0.001*	*0.066 ± 0.014*	*0.043 ± 0.007*	*0.834 ± 0.005*	*0.869 ± 0.005*	*0.841 ± 0.003*
CoRTeC Qwen3-14B	0.054 ± 0.008	0.146 ± 0.006	0.035 ± 0.007	0.048 ± 0.002	0.815 ± 0.009	0.860 ± 0.002	0.835 ± 0.002
maximum-entropy decoder	0.058 ± 0.002	0.145 ± 0.003	0.073 ± 0.008	0.049 ± 0.003	0.818 ± 0.000	0.857 ± 0.001	0.847 ± 0.002
decoder with selection	0.035 ± 0.003	0.120 ± 0.002	0.017 ± 0.002	0.043 ± 0.004	0.822 ± 0.002	0.861 ± 0.004	0.853 ± 0.008
reference AIM	0.052 ± 0.004	0.120 ± 0.003	0.061 ± 0.006	0.048 ± 0.001	0.832 ± 0.011	0.866 ± 0.003	0.839 ± 0.006
auto-configured, no model, with selection	0.020 ± 0.001	0.109 ± 0.001	n/a	n/a	0.830 ± 0.004	0.870 ± 0.006	0.859 ± 0.002
CoRTeC minus decoder	−0.004 [1+/2−] 0.47	+0.001 [2+/1−] 0.88	−0.038 [0+/3−] 0.04	−0.001 [1+/2−] 0.44	−0.003 [1+/2−] 0.63	+0.003 [3+/0−] 0.12	−0.012 [0+/3−] 0.01
CoRTeC minus AIM	+0.002 [2+/1−] 0.80	+0.027 [3+/0−] 0.03	−0.027 [0+/3−] <0.01	−0.000 [2+/1−] 0.86	−0.017 [1+/2−] 0.28	−0.006 [0+/3−] 0.02	−0.004 [1+/2−] 0.42
decoder minus AIM	+0.006 [3+/0−] 0.18	+0.026 [3+/0−] 0.01	+0.012 [3+/0−] 0.27	+0.001 [2+/1−] 0.81	−0.014 [0+/3−] 0.18	−0.009 [0+/3−] 0.05	+0.008 [3+/0−] 0.13
selection minus decoder	−0.023 [0+/3−] 0.01	−0.025 [0+/3−] 0.01	−0.056 [0+/3−] 0.01	−0.006 [0+/3−] 0.20	+0.003 [3+/0−] 0.17	+0.005 [3+/0−] 0.19	+0.006 [3+/0−] 0.31

Adult, n = 300	1-way TV	2-way TV	cond. seen	cond. held-out	TSTR-LR	TSTR-RF	TSTR-GBM
selection minus AIM	−0.017 [0+/3−] 0.03	+0.001 [2+/1−] 0.76	−0.045 [0+/3−] <0.01	−0.005 [1+/2−] 0.23	−0.010 [0+/3−] 0.20	−0.004 [0+/3−] 0.01	+0.014 [3+/0−] 0.01
auto-configured minus AIM	−0.032 [0+/3−] 0.01	−0.011 [0+/3−] 0.05	n/a	n/a	−0.002 [2+/1−] 0.86	+0.004 [2+/1−] 0.28	+0.020 [3+/0−] 0.03
auto-configured minus selection	−0.015 [0+/3−] 0.01	−0.012 [0+/3−] 0.02	n/a	n/a	+0.009 [3+/0−] 0.13	+0.008 [3+/0−] 0.08	+0.006 [2+/1−] 0.28

Adult, n = 1,000	1-way TV	2-way TV	cond. seen	cond. held-out	TSTR-LR	TSTR-RF	TSTR-GBM
real sample (floor)	*0.024 ± 0.002*	*0.061 ± 0.002*	*0.035 ± 0.006*	*0.025 ± 0.005*	*0.849 ± 0.006*	*0.890 ± 0.005*	*0.885 ± 0.005*
CoRTeC Qwen3-14B	0.039 ± 0.004	0.112 ± 0.005	0.027 ± 0.006	0.033 ± 0.003	0.829 ± 0.002	0.863 ± 0.004	0.860 ± 0.003
maximum-entropy decoder	0.041 ± 0.001	0.107 ± 0.001	0.033 ± 0.003	0.042 ± 0.003	0.825 ± 0.001	0.868 ± 0.003	0.863 ± 0.003
decoder with selection	0.027 ± 0.001	0.090 ± 0.001	0.007 ± 0.001	0.033 ± 0.004	0.832 ± 0.004	0.871 ± 0.003	0.871 ± 0.002
reference AIM	0.031 ± 0.001	0.074 ± 0.001	0.034 ± 0.005	0.031 ± 0.001	0.847 ± 0.007	0.882 ± 0.005	0.876 ± 0.006
auto-configured, no model, with selection	0.016 ± 0.000	0.082 ± 0.002	n/a	n/a	0.840 ± 0.006	0.879 ± 0.009	0.881 ± 0.006
CoRTeC minus decoder	−0.002 [1+/2−] 0.52	+0.005 [3+/0−] 0.27	−0.005 [1+/2−] 0.38	−0.009 [0+/3−] 0.07	+0.004 [3+/0−] 0.13	−0.005 [0+/3−] 0.20	−0.002 [0+/3−] 0.25
CoRTeC minus AIM	+0.009 [3+/0−] 0.04	+0.038 [3+/0−] <0.01	−0.007 [0+/3−] 0.09	+0.002 [2+/1−] 0.36	−0.017 [0+/3−] 0.07	−0.019 [0+/3−] 0.07	−0.016 [0+/3−] 0.08
decoder minus AIM	+0.011 [3+/0−] <0.01	+0.033 [3+/0−] <0.01	−0.002 [1+/2−] 0.69	+0.011 [3+/0−] 0.03	−0.021 [0+/3−] 0.03	−0.014 [0+/3−] 0.04	−0.014 [0+/3−] 0.07
selection minus decoder	−0.014 [0+/3−] <0.01	−0.017 [0+/3−] <0.01	−0.025 [0+/3−] <0.01	−0.009 [0+/3−] 0.03	+0.007 [3+/0−] 0.05	+0.003 [3+/0−] 0.01	+0.009 [3+/0−] 0.07
selection minus AIM	−0.003 [0+/3−] 0.05	+0.017 [3+/0−] <0.01	−0.027 [0+/3−] 0.02	+0.002 [2+/1−] 0.60	−0.014 [0+/3−] 0.02	−0.011 [0+/3−] 0.06	−0.005 [0+/3−] 0.16
auto-configured minus AIM	−0.015 [0+/3−] <0.01	+0.009 [3+/0−] <0.01	n/a	n/a	−0.007 [1+/2−] 0.36	−0.003 [1+/2−] 0.56	+0.004 [3+/0−] 0.35
auto-configured minus selection	−0.011 [0+/3−] <0.01	−0.008 [0+/3−] 0.03	n/a	n/a	+0.008 [3+/0−] 0.23	+0.007 [3+/0−] 0.17	+0.009 [3+/0−] 0.11

NHANES, n = 300	1-way TV	2-way TV	cond. seen	cond. held-out	TSTR-LR	TSTR-RF	TSTR-GBM
real sample (floor)	*0.037 ± 0.006*	*0.086 ± 0.006*	*0.021 ± 0.007*	*0.032 ± 0.004*	*0.766 ± 0.009*	*0.740 ± 0.010*	*0.721 ± 0.006*
CoRTeC Qwen3-14B	0.044 ± 0.013	0.161 ± 0.011	0.021 ± 0.013	0.057 ± 0.020	0.710 ± 0.033	0.689 ± 0.028	0.667 ± 0.029
maximum-entropy decoder	0.041 ± 0.001	0.117 ± 0.001	0.024 ± 0.001	0.040 ± 0.002	0.717 ± 0.016	0.713 ± 0.023	0.705 ± 0.015
reference AIM	0.042 ± 0.001	0.103 ± 0.003	0.019 ± 0.003	0.041 ± 0.007	0.752 ± 0.005	0.725 ± 0.004	0.706 ± 0.005
CoRTeC minus decoder	+0.003 [2+/1−] 0.78	+0.044 [3+/0−] 0.02	−0.002 [1+/2−] 0.77	+0.017 [2+/1−] 0.26	−0.007 [1+/2−] 0.75	−0.024 [1+/2−] 0.21	−0.037 [0+/3−] 0.17
CoRTeC minus AIM	+0.001 [2+/1−] 0.87	+0.058 [3+/0−] 0.02	+0.002 [1+/2−] 0.83	+0.016 [2+/1−] 0.37	−0.042 [0+/3−] 0.12	−0.036 [0+/3−] 0.13	−0.039 [0+/3−] 0.18
decoder minus AIM	−0.001 [1+/2−] 0.22	+0.014 [3+/0−] 0.01	+0.004 [3+/0−] 0.15	−0.001 [1+/2−] 0.73	−0.035 [0+/3−] 0.06	−0.012 [1+/2−] 0.45	−0.002 [2+/1−] 0.87

NHANES, n = 1,000	1-way TV	2-way TV	cond. seen	cond. held-out	TSTR-LR	TSTR-RF	TSTR-GBM
real sample (floor)	*0.018 ± 0.002*	*0.040 ± 0.001*	*0.011 ± 0.003*	*0.015 ± 0.001*	*0.785 ± 0.006*	*0.758 ± 0.006*	*0.749 ± 0.005*
CoRTeC Qwen3-14B	0.040 ± 0.013	0.151 ± 0.008	0.022 ± 0.008	0.054 ± 0.016	0.732 ± 0.031	0.686 ± 0.037	0.669 ± 0.034
maximum-entropy decoder	0.031 ± 0.002	0.087 ± 0.002	0.018 ± 0.003	0.026 ± 0.003	0.769 ± 0.011	0.745 ± 0.023	0.722 ± 0.032
reference AIM	0.030 ± 0.002	0.071 ± 0.001	0.022 ± 0.007	0.035 ± 0.016	0.734 ± 0.027	0.701 ± 0.033	0.676 ± 0.022
CoRTeC minus decoder	+0.009 [2+/1−] 0.34	+0.064 [3+/0−] <0.01	+0.004 [3+/0−] 0.25	+0.028 [3+/0−] 0.09	−0.037 [0+/3−] 0.10	−0.059 [0+/3−] 0.06	−0.053 [0+/3−] 0.08
CoRTeC minus AIM	+0.009 [2+/1−] 0.33	+0.080 [3+/0−] <0.01	−0.000 [2+/1−] 0.96	+0.018 [3+/0−] 0.21	−0.002 [2+/1−] 0.96	−0.015 [1+/2−] 0.73	−0.007 [1+/2−] 0.81
decoder minus AIM	+0.000 [2+/1−] 0.85	+0.016 [3+/0−] 0.01	−0.004 [1+/2−] 0.34	−0.010 [1+/2−] 0.47	+0.035 [3+/0−] 0.24	+0.044 [3+/0−] 0.21	+0.046 [3+/0−] 0.07

E.3 Per-pair reading. For every pair of columns we compared the pairwise total variation of the model's draws and of the decoder's draws against the private table, on the seed-42 split. On Adult the model is below the decoder on the pairs it recognises as logically linked. For education with education number, the decoder reads 0.26 to 0.40 and the model 0.08 to 0.32 across backbones and sizes. For marital status with relationship the model reads 0.08 to 0.30 against the decoder's 0.33 to 0.40, and for relationship with sex 0.03 to 0.19 against 0.22 to 0.26. The same holds for workclass with occupation. On the pairs the release describes only through the cohorts, the model is at or above the decoder. On bank marketing the model repairs days since contact with the previous outcome. It raises the error on education with contact, contact with the previous outcome, marital status with education, and loan with contact, four nearly independent pairs on which the decoder matches the real sample. The same pattern appears at both sizes and on every backbone.

E.4 The backbone ladder. One Adult release and one NHANES release, decoded with the same prompts, settings and evaluator, with only the model changed. Each cell gives single-column and pairwise total variation, then the three students.

backbone	Adult n = 300	utility	Adult n = 1,000	utility	NHANES n = 300	utility	NHANES n = 1,000	utility
Qwen2.5-7B	0.110 / 0.220	0.81 / 0.85 / 0.83	0.089 / 0.178	0.81 / 0.86 / 0.85	0.045 / 0.160	0.64 / 0.64 / 0.61	0.039 / 0.145	0.66 / 0.64 / 0.62
Qwen3-8B	0.068 / 0.164	0.80 / 0.85 / 0.82	0.048 / 0.124 (2 draws)	0.82 / 0.85 / 0.84	0.042 / 0.181	0.65 / 0.65 / 0.66	0.034 / 0.156	0.75 / 0.69 / 0.67
Qwen3-14B	0.046 / 0.139	0.81 / 0.86 / 0.83	0.036 / 0.108	0.83 / 0.86 / 0.86	0.029 / 0.158	0.68 / 0.66 / 0.65	0.027 / 0.150	0.70 / 0.65 / 0.64
Qwen3-32B (AWQ)	0.049 / 0.138	0.82 / 0.86 / 0.84	0.035 / 0.103	0.82 / 0.86 / 0.85	0.047 / 0.157	0.68 / 0.68 / 0.66	0.036 / 0.123	0.75 / 0.73 / 0.70
Gemini 3.6 Flash	0.027 / 0.110	0.81 / 0.86 / 0.84	not run		not run		not run	
maximum-entropy decoder	*0.058 / 0.146*	*0.82 / 0.86 / 0.85*	*0.041 / 0.106*	*0.83 / 0.87 / 0.86*	*0.042 / 0.117*	*0.71 / 0.69 / 0.69*	*0.031 / 0.088*	*0.76 / 0.72 / 0.69*
reference AIM	*0.056 / 0.122*	*0.85 / 0.87 / 0.85*	*0.029 / 0.072*	*0.85 / 0.89 / 0.88*	*0.042 / 0.100*	*0.75 / 0.72 / 0.71*	*0.029 / 0.070*	*0.75 / 0.71 / 0.66*
real sample	*0.043 / 0.107*	*0.83 / 0.87 / 0.84*	*0.026 / 0.061*	*0.86 / 0.89 / 0.88*	*0.034 / 0.081*	*0.77 / 0.73 / 0.72*	*0.016 / 0.039*	*0.78 / 0.75 / 0.74*

E.5 Bank marketing at ε = 2, one split, three draws. The held-out conditional family is not defined on this schema.

condition	n	1-way TV	2-way TV	cond. seen	TSTR-LR	TSTR-RF	TSTR-GBM
real sample	300	*0.025*	*0.052*	*0.014*	*0.833*	*0.849*	*0.833*
permuted target	300	*0.027*	*0.056*	*0.096*	*0.415*	*0.467*	*0.463*
CoRTeC, Qwen3-14B	300	0.040	0.112	0.013	0.802	0.799	0.781
maximum-entropy decoder	300	0.026	0.066	0.036	0.824	0.802	0.802
reference AIM	300	0.034	0.068	0.021	0.815	0.805	0.772
real sample	1,000	*0.014*	*0.028*	*0.008*	*0.862*	*0.871*	*0.860*
permuted target	1,000	*0.013*	*0.031*	*0.089*	*0.503*	*0.483*	*0.485*
CoRTeC, Qwen3-14B	1,000	0.034	0.101	0.011	0.810	0.784	0.759
maximum-entropy decoder	1,000	0.013	0.043	0.013	0.837	0.848	0.837
reference AIM	1,000	0.024	0.049	0.006	0.839	0.824	0.805
real sample of 4,000 rows (utility ceiling)		*0.007*	*0.014*	*0.005*	*0.869*	*0.890*	*0.891*

E.6 The three papers' own protocols on Adult, ε = 1, one split. The published rows are the papers' own. Our rows are scored by our implementation of each paper's metrics. Under the DP-LLMTGen protocol the distance columns are on our scorer's scale, which reads the training set itself at 0.009 against the paper's 0.004, so only the utility columns can be compared across the two sources. Under the Tab-PE protocol, the embedding-space precision and recall depend on the implementation and are left out. Rows marked three seeds are means over the three Tab-PE splits.

protocol and method	fidelity (the paper's measure) ↓	utility (the paper's measure) ↑
DP-LLMTGen protocol (1-way, 2-way TVD against the test split; XGBoost ACC, AUC)		
DP-LLMTGen, Llama-2 7B (published)	0.038, 0.068	0.831, 0.879
RAP++ (published)	0.054, 0.099	0.803, 0.807
CoRTeC, Qwen2.5-7B (ours)	0.060, 0.121	0.831, 0.881
maximum-entropy decoder (ours)	0.034, 0.089	0.818, 0.888
reference AIM (ours)	0.018, 0.046	0.850, 0.902
reference MST (ours)	0.018, 0.061	0.813, 0.836
DP fine-tuned Llama-2 7B, our run of the procedure	0.103, 0.177	0.795, 0.841
DP-2Stage protocol (their split and metric suite; Pair, HIST, CorAcc in %; F1, AUC in %)		
real data (published)	97.5, 99.1, 97.3	69.9, 91.7
AIM (published)	77.1, 88.4, 86.4	59.6, 86.8
DP-2Stage-O (published)	72.3, 88.5, 55.4	30.4, 61.6
DP-2Stage-O, our run of their code	63.0, 84.7, 56.4	26.2, 59.9
CoRTeC, Qwen2.5-7B (ours)	70.8, 84.6, 63.6	61.8, 87.1
maximum-entropy decoder (ours)	74.3, 87.6, 69.8	62.8, 87.1
reference AIM (ours)	93.1, 97.2, 83.1	62.9, 87.7
reference MST (ours)	92.6, 97.2, 74.2	51.5, 84.0
Tab-PE protocol (1-WD, 2-WD against the private table; TabICL accuracy, macro F1)		
real data (published UB)	0.014, 0.027	84.41, 75.68
AIM (published)	0.017, 0.034	83.36, 76.10
Tab-PE (published)	0.049, 0.086	82.22, 73.66
Tab-PE, our run of their code, three seeds	0.044, 0.093	82.19, 72.96
CoRTeC, Qwen3-14B, three seeds	0.040, 0.100	81.44, 74.46
CoRTeC, Qwen3-32B, three seeds	0.039, 0.101	81.42, 74.98
CoRTeC, Qwen2.5-7B, three seeds	0.088, 0.176	81.75, 74.10
maximum-entropy decoder, three seeds	0.045, 0.109	81.97, 76.55
reference AIM, three seeds	0.022, 0.061	81.69, 70.84
reference MST, three seeds	0.023, 0.062	80.86, 64.68

E.7 Other budgets. The reference AIM and MST at $\varepsilon = 0.5$ and 1 under our protocol (five draws on Adult and five on NHANES), with the Qwen2.5-7B CoRTeC arm and the release's own maximum-entropy decoder at the same budgets (three draws; the decoder's $\varepsilon = 2$ rows repeat Table 3), and the DP-LLMTGen protocol at $\varepsilon = 0.5$. The $\varepsilon = 2$ rows of Table 3 are the reference. At $\varepsilon = 0.5$ and 1 the decoder improves on the 7B arm on single columns, pairs and every student at both sizes, so that arm's errors come from the backbone and not from the release. The reference AIM fits the private table more closely than the decoder at every budget. At $n = 300$ and $\varepsilon = 0.5$ the decoder's students score higher (0.819, 0.855 and 0.850 against 0.811, 0.814 and 0.787, one split).

Adult, $n = 300$	ε	1-way TV	2-way TV	cond. seen	cond. held-out	TSTR-LR	TSTR-RF	TSTR-GBM
reference AIM	0.5	0.050	0.126	0.061	0.084	0.811	0.814	0.787
reference AIM	1	0.054	0.125	0.078	0.051	0.823	0.859	0.830
reference MST	0.5	0.053	0.129	0.132	0.054	0.774	0.810	0.780
reference MST	1	0.051	0.127	0.144	0.042	0.741	0.798	0.754
CoRTeC, Qwen2.5-7B	0.5	0.127	0.255	0.066	0.060	0.754	0.826	0.809
CoRTeC, Qwen2.5-7B	1	0.115	0.223	0.122	0.067	0.763	0.826	0.798
CoRTeC, Qwen2.5-7B	2	0.110	0.220	0.090	0.054	0.813	0.852	0.830
maximum-entropy decoder	0.5	0.086	0.190	0.074	0.071	0.819	0.855	0.850
maximum-entropy decoder	1	0.066	0.157	0.069	0.060	0.821	0.859	0.857
maximum-entropy decoder	2	0.058	0.146	0.082	0.049	0.818	0.856	0.847

Adult, n = 300	ε	1-way TV	2-way TV	cond. seen	cond. held-out	TSTR-LR	TSTR-RF	TSTR-GBM
Adult, n = 1,000								
reference AIM	0.5	0.037	0.087	0.037	0.031	0.841	0.880	0.873
reference AIM	1	0.033	0.080	0.051	0.032	0.837	0.873	0.861
reference MST	0.5	0.032	0.089	0.135	0.037	0.758	0.806	0.791
reference MST	1	0.032	0.086	0.136	0.044	0.781	0.817	0.802
CoRTeC, Qwen2.5-7B	0.5	0.107	0.208	0.052	0.046	0.808	0.863	0.848
CoRTeC, Qwen2.5-7B	1	0.095	0.183	0.078	0.044	0.807	0.853	0.840
CoRTeC, Qwen2.5-7B	2	0.089	0.178	0.061	0.039	0.808	0.855	0.849
maximum-entropy decoder	0.5	0.078	0.164	0.034	0.062	0.817	0.864	0.854
maximum-entropy decoder	1	0.052	0.123	0.036	0.048	0.819	0.867	0.857
maximum-entropy decoder	2	0.041	0.106	0.034	0.045	0.826	0.870	0.863
NHANES, n = 300								
reference AIM	0.5	0.053	0.122	0.025	0.046	0.716	0.681	0.665
reference AIM	1	0.040	0.104	0.040	0.042	0.716	0.709	0.693
reference MST	0.5	0.050	0.126	0.033	0.063	0.704	0.655	0.642
reference MST	1	0.045	0.109	0.027	0.046	0.703	0.658	0.622
Tab-PE (1,000-row population)	0.5	0.163	0.280	0.028	0.048	0.702	0.674	0.667
Tab-PE (1,000-row population)	1	0.098	0.184	0.053	0.049	0.701	0.682	0.653

DP-LLMTGen protocol, ε = 0.5	1-way	2-way	3-way	4-way	5-way	ACC	AUC
DP-LLMTGen (published)	0.058	0.095	0.126	0.154	0.183	0.833	0.887
RAP++ (published)	0.066	0.116	0.162	0.208	0.251	0.766	0.758
CoRTeC, Qwen2.5-7B (ours)	0.075	0.145	0.225	0.320	0.424	0.833	0.885
maximum-entropy decoder (ours)	0.049	0.112	0.193	0.293	0.409	0.819	0.882
reference AIM (ours)	0.021	0.057	0.106	0.173	0.259	0.821	0.845
reference MST (ours)	0.020	0.061	0.117	0.190	0.281	0.808	0.823
DP fine-tuned Llama-2 7B, our run	0.103	0.181	0.253	0.329	0.412	0.772	0.815
training set, our scorer	*0.009*	*0.025*	*0.057*	*0.112*	*0.190*	*0.877*	*0.931*

The published Tab-PE row's embedding-space precision and recall are 34.3 and 77.3. Our Tab-PE rerun reads 66.7 and 74.6, and our CoRTeC rows 61 to 72 and 71 to 84, with a 64-16-64 autoencoder trained for each run. The precision column therefore depends on the implementation and is not compared.

E.8 Stage C on NHANES. We use the rule the implementation ships. The tolerance is derived from the release (0.107 for the auto-configured release), and no verdict is given while a released cell holds fewer than 20 synthetic rows. Under that rule no single 300-row draw receives a verdict, because two or three of the four released cells are thin. The three draws pooled (900 rows) are within the bound at 0.087, against the tolerance of 0.107. The real-sample ceiling passes at 0.088 and the permuted floor fails at 0.139, so the verdict holds at 95 % simultaneous confidence over the four cells. Stage C cost ε = 1 on top of the release's ε = 2.

Appendix F. Membership Inference, Distance to Closest Record and Memorisation

F.1 Membership inference. We use the standard membership game [33] in the form Stadler et al. apply to synthetic data [34]. There are 3,000 candidates, half from the training split and half held out. The adversary sees the three draws of n = 300 (900 rows) and must say whether each candidate was in the training set. We run four attacks: nearest-neighbour distance, a discriminative shadow model, a per-record likelihood-ratio test in the style of Carlini et al. [11] over sixteen bootstrap density models, and exact matching. The advantage is $2 \cdot \mathrm{AUC} - 1$. The positive control passes 900 real training rows off as synthetic. On Adult the 1,500 member candidates are drawn from a training split of 26,048 rows, so only a few of them can be among those 900 rows, and the control's exact-match advantage is small, 0.098 as measured. On NHANES, whose training split has 2,999 rows, it is 0.48. The other three attacks separate the control on both datasets.

arm, ε = 2	dataset	strongest advantage (attack)	nearest-neighbour AUC (95 %)	exact matches
CoRTeC, Qwen3-14B	Adult	0.022 (shadow model)	0.491 (0.477 to 0.506)	0
CoRTeC, Qwen3-32B (AWQ)	Adult	0.035 (likelihood ratio)	0.492 (0.478 to 0.507)	0
CoRTeC, Qwen3-8B	Adult	0.026 (nearest neighbour)	0.487 (0.473 to 0.502)	0
CoRTeC, Gemini 3.6 Flash	Adult	0.022 (nearest neighbour)	0.489 (0.475 to 0.503)	0
CoRTeC, anonymised, Gemini 3.6 Flash	Adult	0.021 (nearest neighbour)	0.489 (0.475 to 0.504)	0
maximum-entropy decoder	Adult	0.025 (nearest neighbour)	0.488 (0.473 to 0.503)	0
maximum-entropy decoder with the selection step	Adult	0.024 (likelihood ratio)	0.493 (0.478 to 0.508)	0
auto-configured release, no model, with selection	Adult	0.025 (likelihood ratio)	0.488 (0.475 to 0.503)	0
one-shot from the column names (no release)	Adult	0.012 (nearest neighbour)	0.494 (0.480 to 0.510)	5
reference AIM	Adult	0.024 (nearest neighbour)	0.488 (0.474 to 0.503)	0
reference MST	Adult	0.032 (nearest neighbour)	0.484 (0.470 to 0.499)	0
DP-2Stage, GPT-2	Adult	0.021 (shadow model)	0.493 (0.478 to 0.507)	0
DP fine-tuned Qwen3-14B	Adult	0.024 (likelihood ratio)	0.493 (0.479 to 0.508)	0
CoRTeC, Qwen3-14B	NHANES	0.026 (likelihood ratio)	0.509 (0.478 to 0.537)	0
CoRTeC, Qwen3-32B (AWQ)	NHANES	0.049 (likelihood ratio)	0.516 (0.486 to 0.545)	0
maximum-entropy decoder	NHANES	0.044 (likelihood ratio)	0.503 (0.474 to 0.533)	0
reference AIM	NHANES	0.048 (shadow model)	0.506 (0.478 to 0.535)	0
reference MST	NHANES	0.039 (shadow model)	0.504 (0.473 to 0.533)	0
DP fine-tuned Qwen3-14B	NHANES	0.060 (likelihood ratio)	0.473 (0.445 to 0.501)	0
Tab-PE	NHANES	0.037 (shadow model)	0.511 (0.480 to 0.541)	0
permitted advantage at ε = 2		*0.762*		

Every nearest-neighbour interval contains chance. These attacks bound leakage only as far as they are strong. A black-box audit of the kind Annamalai and De Cristofaro give for DP-SGD [2] would return an empirical ε instead of a null result, and is the natural next measurement. On Adult the nearest-neighbour statistic reads slightly below chance for every method, CoRTeC, the decoder, AIM and MST alike. The private rows lie slightly farther from the synthetic rows than held-out rows do, which is a property of the split and the statistic, not of any method.

F.2 Distance to the closest record. For every synthetic row we compute the L1 distance to the nearest training row over the 15 Adult columns. Numerical columns are scaled to [0, 1] on the training split, and a categorical column counts 1 when it does not match. The reference curve is the real test split measured against the training split. Figure 4 shows the Adult arms at ε = 2. No arm contains an exact copy of a training row, while the real test split itself contains 0.11 % exact duplicates. The medians are 0.145 for real held-out rows, 0.185 for the Gemini arm, 0.19 for the reference AIM, 0.34 for MST, 0.38 and 0.44 for the Qwen3-14B arm at the two sizes, 1.18 for the decoder, 1.8 for DP-2Stage, 3.6 for Tab-PE and 0.105 for the DP fine-tuned Qwen3-14B. The last is the smallest, so we ran a paired check. We compared each row's distance to a random 6,513-row subset of the training split with its distance to the 6,513-row test split. A generator that copies training rows sits closer to the first, and one that produces typical rows sits equally close to both. The share of rows closer to the training subset is between 0.47 and 0.54 for every arm, against 0.50 expected, so the fine-tuned model's small distances reflect typical rows, not copying.

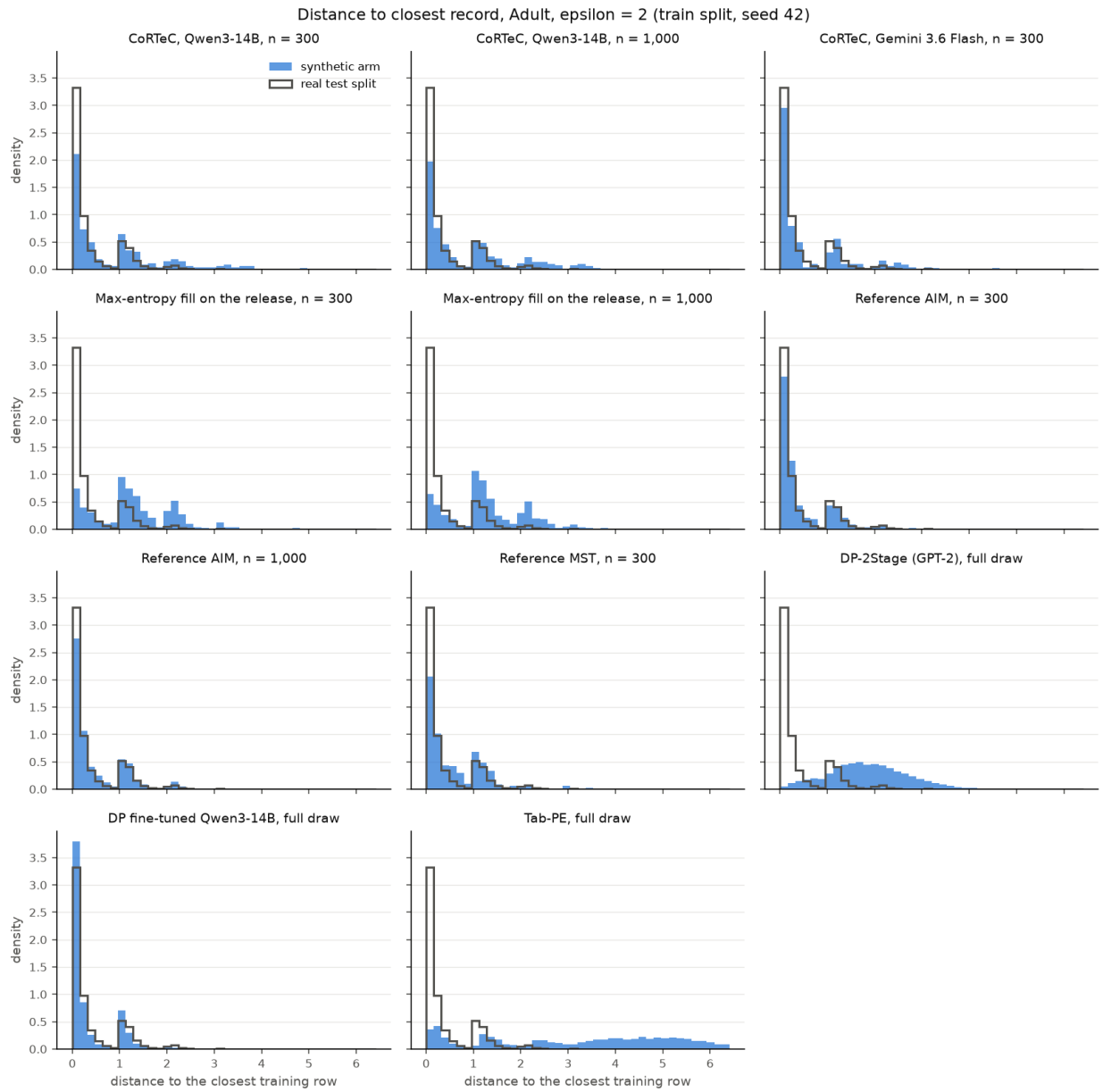


Figure 4: Distance to the closest training row, Adult at ε = 2, for each arm against the curve of the real test split.

F.3 Memorisation tests of the served backbone. We ran the tests of Bordt, Nori and Caruana [4] (package tabmemcheck 0.1.6) on the Adult training file against gemini-3.6-flash, with reasoning off and temperature 0, in 304 calls. In the header test, given the header and the first characters of row 1, the model reproduced data rows 2 to 25 of the file verbatim, including the final-weight values. The opening of the Adult file appears in countless tutorials, and the model has memorised it, as prior work found for GPT-4. Row completion at random positions, with ten real rows as the prefix, succeeded 0 times in 25. Completion of the nearly unique final-weight column from the other fourteen succeeded 0 times in 25. The first-token test scored 8 of 25, against 7 of 25 for always choosing the most common token. In feature completion, each categorical column was predicted from the other columns for 25 rows with five examples. Every column was completed at or below the rate of a gradient-boosting classifier trained on the other columns: education 25 of 25 against a predictability of 1.00, occupation 7 of 25 against 0.39, workclass 14 of 25 against 0.76, relationship 22 of 25 against 0.79, and income 21 of 25 against 0.87. Every column was above the majority choice, which reflects the knowledge of the column meanings that the anonymised control removes.

None of the file's first fifty rows appears in any CoRTeC draw on Gemini, named or anonymised, or on Qwen3-14B or 32B at either size. The one-shot control of Table 3, which has no release, shows that recitation in a draw. Of its rows, 208 to 211 of 300 and 697 to 703 of 1,000 are verbatim copies of the file's first 25 rows (22 to 24 distinct rows, each repeated about nine and thirty times). Its three $n = 300$ draws hold five exact matches among the 3,000

membership candidates, three members and two held out, which no attack turns into an advantage (0.012). Against the whole public file of 32,561 rows, no draw of any CoRTeC arm, of the decoder with or without selection, or of the reference AIM holds a verbatim row. Rows that agree on the fourteen columns other than the nearly unique final weight occur in the CoRTeC draws at the reference AIM's rate (10 to 16 % against 8 to 14 % of a draw at $n = 300$). That is coincidence on a table with many near-duplicate rows, not copying. The Qwen3 backbones were not given these tests.

Appendix G. An Indicative Summary in Percent

For a reader who wants one number per axis, we scale fidelity against a real sample of the same size and utility between the no-information floor and the ceiling. Fidelity % is 100 times the mean, over the four fidelity measures, of min(1, real-sample error / method error). So 100 means as close to the private table as a real sample of that size, and 50 means twice as far. Utility % is 100 times the mean, over the three students, of (method AUC − permuted-target AUC) / (ceiling AUC − permuted-target AUC), clipped to [0, 100]. Each row averages every dataset, n and seed the arm was run on at $\varepsilon = 2$, and the cells column counts them. The summary is indicative only and is not a ranking. The tables above are the evidence. Rows are ordered by the number of cells and then by fidelity. Arms with one or two cells are single-split signals. The auto-configured arm's fidelity mean uses the two marginal measures only, because its conditional cells are not comparable.

method	cells	fidelity %	utility %
reference AIM	14	78	85
maximum-entropy decoder of the release	14	72	86
CoRTeC, Qwen3-14B	14	68	79
auto-configured release, no model, with selection	6	93	93
maximum-entropy decoder with the selection step	6	89	91
DP fine-tuned Qwen3-14B	6	27	63
DP fine-tuned Qwen2.5-7B	6	26	61
Tab-PE	5	17	84
CoRTeC, Qwen3-32B (AWQ)	4	66	80
CoRTeC, Qwen2.5-7B	4	49	67
DP fine-tuned Llama-2 7B	4	21	71
DP-2Stage, GPT-2	4	19	15
CoRTeC, Qwen3-8B	2	71	83
one-shot from the column names (no release)	2	23	85
CoRTeC, Gemini 3.6 Flash	1	96	86